The Fearless Premium: Workplace Psychological Safety and Stock Returns

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ABSTRACT

Organizational research identifies psychological safety—the shared belief that employees can speak up with ideas, questions, or mistakes without fear of negative consequences—as a critical driver of learning, innovation, and team performance. Despite its prominence in organizational science, its implications for firm value and asset prices remain largely unexplored. We construct a novel firm-level measure of psychological safety (PS) from Glassdoor employee reviews using a fine-tuned GPT model. Our measure captures nuanced aspects of workplace voice and interpersonal risk that are not reflected in existing corporate culture metrics. We show that PS predicts future innovation but is not associated with higher contemporaneous market valuations. In contrast, an equal-weighted long–short portfolio that buys firms with high PS and shorts firms with low PS earns an annualized Fama–French six-factor alpha of 5.18%. These returns are robust to alternative weighting schemes, other factor models, and controls for intangible-related factors. Cross-sectional Fama–MacBeth tests show that the effect of PS on stock returns is concentrated among innovation-intensive firms and in the component of PS measured from white-collar employee reviews. Firms with higher PS also experience systematically positive earnings surprises and stronger announcement returns, consistent with the gradual incorporation of its value into prices. Overall, our findings suggest that psychological safety is an economically meaningful yet underappreciated organizational attribute in financial markets.

Keywords: Corporate culture, employee reviews, innovation, textual analysis, LLM, GPT, machine learning, stock returns.

JEL Classification: G12, G14, G30, O31, J24, M14.

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1 Introduction

In the organizational literature, psychological safety is defined as a shared belief that the team is safe for interpersonal risk-taking, referring to a team-level climate in which individuals feel able to speak up, ask questions, admit mistakes, or challenge prevailing views without fear of embarrassment, rejection, or punishment (Edmondson, 1999).

This idea appears prominently in both accounts of exceptional team performance and analyses of organizational failure. In a widely read New York Times account of Google's Project Aristotle, psychological safety was described as the single most important factor distinguishing Google's highest-performing teams (Duhigg, 2016). Similar lessons emerge from investigations of the Space Shuttle Challenger disaster. Post-accident analyses show that critical technical concerns had been raised before launch but were not forcefully escalated or acted upon in an organizational context marked by norms that discouraged dissent, promoted deference to hierarchy, and emphasized schedule adherence over open challenge (Vaughan, 1996). In its Lessons Learned from Challenger safety message[1], NASA identifies as a central lesson that "we must allow people to come forward with their concerns without fear of repercussion", underscoring how inhibited voice and fear of speaking up can have serious consequences.

Together, these cases illustrate a broader insight from organizational practice: performance depends not only on technical expertise or individual talent, but also on whether individuals feel able to raise concerns, admit mistakes, and challenge prevailing views without fear of negative consequences (Edmondson, 2018). Importantly, psychological safety is conceptually distinct from having a "strong" corporate culture or from other positive cultural attributes. A firm may possess clearly articulated values, strong cohesion, or high performance standards, yet still exhibit low psychological safety if employees fear embarrassment, exclusion, or retaliation for speaking up (Edmondson and Lei, 2014). Similarly, prior research emphasizes that psychological safety is not synonymous with trust, cohesiveness, or simply "being nice," nor does it imply the absence of accountability or performance pressure (Edmondson, 1999, 2018). Rather, it specifically concerns the perceived interpersonal costs of voice.

While organizational research shows that psychological safety can foster team learning, knowledge sharing, and innovation—based largely on survey-based and qualitative evidence at the team or project level (Edmondson, 1999, 2018)—much less is known about whether such organizational climates are reflected in capital market outcomes. Psychological safety is inherently intangible and difficult to observe externally, raising the question of whether investors recognize and incorporate it into firm valuation.

[1] NASA, "Lessons Learned from Challenger" Safety Message, January 2021.

To study these questions, we develop a novel firm-level measure of psychological safety (PS) using textual employee reviews from Glassdoor over 2011–2021. Methodologically, our approach follows an LLM-labeled supervised-learning design (Pangakis and Wolken, 2024): we first use a GPT-style generative language model to generate theory-guided psychological safety annotations at the review level and then use these annotations to fine-tune a BERT-style transformer classifier (He et al., 2023) that can be applied at scale.[2] This approach allows us to leverage the contextual understanding of frontier LLMs while retaining the scalability and computational efficiency of a dedicated classifier. As a result, our measure captures subtle expressions of voice, feedback, and error tolerance that are not readily observable in Glassdoor's numerical ratings and that, while correlated as expected, are distinct from broad measures of corporate culture (Li et al., 2020) and employee satisfaction (Edmans, 2011; Boustanifar and Kang, 2022).

To better understand what our measure captures, we first examine how PS correlates with firm characteristics and existing measures of corporate culture. We find that PS is largely unrelated to firm size and profitability, but positively associated with growth opportunities and, to a lesser extent, intangible capital intensity. The absence of a relation with profitability suggests that PS does not merely proxy for contemporaneously "better-performing" firms, but instead reflects an aspect of interpersonal culture that is distinct from current financial success. When compared to the culture measures of Li et al. (2020), PS is positively correlated with dimensions such as teamwork and innovation, yet exhibits weak or even negative correlations with integrity and quality. Firms included in the Fortune 100 Best Companies to Work For list (Best Workplaces, hereafter) tend to exhibit higher PS. Importantly, observable firm characteristics, corporate culture measures, and industry and year fixed effects together explain only about 5% of the cross-sectional variation in PS. These patterns suggest that PS is related to collaborative and idea-oriented aspects of corporate culture—providing face validity for our measure—while remaining empirically distinct from broader or more formalized cultural attributes, consistent with the conceptual distinction emphasized in prior research (Edmondson, 1999, 2018).

We further benchmark PS against employee ratings from Glassdoor. PS is positively correlated with overall firm ratings and with perceived culture and work–life balance, but is weakly or negatively related to compensation and largely unrelated to career opportunities or diversity ratings. Even after accounting for these ratings, the majority of the variation in

[2]This use of LLMs as annotators is consistent with evidence that they can perform comparably to, and in some settings better than, human annotators (Gilardi et al., 2023; Törnberg, 2023; Aguda et al., 2024; Asirvatham et al., 2026). We provide additional validation in our setting by manually labeling an independent sample of employee reviews and showing that the resulting human labels are highly consistent with the model-generated classifications.

PS remains unexplained, indicating that our measure captures a distinct dimension of the organizational environment that is not subsumed by existing employee-based assessments.

We then study the implications of PS for innovation, market valuation, and stock returns. First, PS predicts future innovation, as measured by patent grants, even after controlling for R&D investment, alternative measures of corporate culture, and employee ratings.[3] Second, we show that PS—unlike general employee satisfaction, as measured by overall Glassdoor ratings—is not associated with higher contemporaneous market valuations. This suggests that the potential value of psychological safety is not fully reflected in prices. Third, firms with higher PS earn significant risk-adjusted excess returns. The annualized Fama–French six-factor alpha is 5.18% (t-statistic = 3.12) in equal-weighted portfolios and 4.26% (t-statistic = 2.73) in value-weighted portfolios. These returns are robust across multiple factor models, including the q-factor model of Hou et al. (2021) (Q5) and the mispricing factor model of Stambaugh and Yuan (2017) (SY4), as well as to controls for intangible-related factors.

One potential concern is that the excess returns associated with PS simply reflect higher general employee satisfaction, as captured by overall Glassdoor ratings. The valuation results discussed above already suggest that this explanation is unlikely, as PS and overall Glassdoor ratings exhibit very different associations with contemporaneous market valuation. We nevertheless address this concern directly in two ways. First, we form portfolios based on overall Glassdoor ratings and find that the corresponding long–short portfolio generates alphas that are small in magnitude and statistically insignificant. Second, we sort firms on residualized PS, defined as the residual from a regression of PS on overall Glassdoor ratings. The resulting long–short portfolios generate statistically significant alphas that are very similar in magnitude to those obtained using the non-residualized measure. These results show that the return premium associated with PS is distinct from the effect of general employee satisfaction.

While portfolio tests have the advantage of producing economically interpretable long–short return spreads and are relatively robust to measurement error and nonlinearities in the pricing relation, they are less suited to estimating the marginal effect of PS conditional on correlated firm characteristics. To complement our time-series portfolio results, we use Fama–MacBeth regressions that allow us to control flexibly for a broad set of standard return predictors, Glassdoor ratings, and the Best Workplaces indicator. They also allow us to test whether the marginal return predictability of PS is stronger in settings where organizational theory predicts it should be most value-relevant. We find a positive but not statistically significant average coefficient on PS in these cross-sectional firm-level Fama–MacBeth regressions, suggesting

[3]By contrast, the coefficient on overall Glassdoor ratings in predicting future patent grants is consistently negative and statistically insignificant.

that its pricing implications are not uniform across firms. Consistent with organizational theory, we further find that return predictability is concentrated among innovation-intensive firms and when PS is measured using reviews from white-collar employees, suggesting that the return premium is strongest where knowledge sharing, experimentation, and employee voice are especially important.

To shed light on the mechanisms underlying the return premium to PS, we examine earnings announcements. We find that firms with higher PS systematically deliver positive earnings surprises and experience stronger market reactions around announcements, suggesting that the value of psychological safety is incorporated into prices gradually rather than being immediately reflected in valuations.

A growing body of research examines the measurement and economic implications of corporate culture for firm outcomes (Li et al., 2020; Graham et al., 2020, 2022; Li et al., 2026).[4] Consistent with the importance of culture emphasized in academic work, 92% of executives surveyed by Graham et al. (2022) believe that improving corporate culture would increase firm value. In their survey of the corporate culture literature, Gorton et al. (2023) conclude that "much remains to be studied and developed on corporate culture," arguing that "the right way to research culture empirically is to unpack it." Indeed, corporate culture encompasses multiple dimensions, each of which may have distinct—and potentially opposing—implications for firm behavior and outcomes.

In line with this perspective, several studies have measured and evaluated specific dimensions of corporate culture, including workplace flexibility (Au et al., 2021), sexual harassment (Au et al., 2024), workers' access to meaningful projects (Tambe et al., 2020), integrity (Guiso et al., 2015), employee satisfaction (Edmans, 2011; Boustanifar and Kang, 2022), culture in family firms (Huang et al., 2015), diversity, equity, and inclusion (Cai et al., 2022; Corritore et al., 2020; Briscoe-Tran, 2023; Edmans et al., 2025), and differences in stakeholders' perspectives on corporate culture (Li et al., 2026). We contribute to this literature by studying an important yet understudied dimension of corporate culture: psychological safety.

Most of the above-mentioned studies do not examine the stock return implications of the cultural dimensions they measure. Among those that do, the evidence is mixed. While Edmans (2011) and Boustanifar and Kang (2022) document excess returns for firms with higher employee satisfaction, Edmans et al. (2024) show that such outperformance is confined to countries with high labor market flexibility, and Green et al. (2019) find no evidence that firms with higher employee satisfaction—measured using Glassdoor ratings—earn excess

[4]Comprehensive surveys of this literature are provided by Grennan and Li (2023) and Gorton et al. (2023).

returns.[5] Similarly, Edmans et al. (2025) find no evidence that firms with higher diversity, equity, and inclusion earn excess returns. Taken together, these findings suggest that different cultural attributes can have distinct economic implications, and that excess returns arise only when markets fail to fully incorporate the value of specific cultural dimensions. We contribute to this literature by constructing a novel measure of psychological safety and showing that firms with higher psychological safety earn economically and statistically significant risk-adjusted returns.

We also contribute to the growing text-as-data literature, particularly studies that use textual data to construct and evaluate the effects of hard-to-measure firm-level variables.[6] More specifically, recent research in accounting and finance demonstrates the usefulness of large language models (LLMs) for extracting economically meaningful but difficult-to-measure information from financial texts (Lopez-Lira and Tang, 2026; Jha et al., 2024; Li et al., 2025; Choi and Kim, 2025; Li et al., 2026). A strand of this work shows that LLMs can extract information from text that is useful for forecasting future outcomes (e.g., Jha et al., 2024; Lopez-Lira and Tang, 2026). Our use of LLMs is more direct: we use ChatGPT to generate theory-guided annotations of employee reviews based on their psychological safety content. This task requires contextual and semantic interpretation of information contained in the review itself, rather than prediction of a future realization, making it particularly well suited to LLM-based annotation. To the best of our knowledge, our work is among the first applications of this LLM-labeled supervised-learning design in the finance literature.

The remainder of the paper is organized as follows. Section 2 describes the dataset and key variables used in the empirical analysis. Section 3 describes the construction of our psychological safety measure. Section 4 presents the empirical results, and Section 5 concludes.

2 Data

Our primary data source is employee review data from Glassdoor.com, one of the largest online platforms on which current and former employees anonymously evaluate their employers.

[5] Green et al. (2019) show that *changes* in employee ratings predict stock returns and argue that this effect reflects employees' information about near-term changes in firms' tangible fundamentals rather than corporate culture or employee satisfaction; related evidence is reported in Sheng (2025).

[6] This now-large literature spans many settings; illustrative recent examples include measures of climate change exposure (Sautner et al., 2023), biodiversity risk (Giglio et al., 2025), intangible talks (Filipović and Wagner, 2026), intangible liabilities (Boustanifar and Verriest, 2026), firms' scale and scope (Hoberg and Phillips, 2025), management clarity (Barth et al., 2023; Boustanifar and Mansouri, 2026), regulatory intensity (Kalmenovitz, 2023), brand value (Boustanifar and Kang, 2024), and financial constraints (Bodnaruk et al., 2015).

Prior research suggests that Glassdoor reviews reflect employees' private information about organizational quality and internal practices and contain economically meaningful information, making them a valuable source for empirical analysis (Bams et al., 2025; Campbell and Shang, 2022; Dube and Zhu, 2021; Lee et al., 2021; Huang et al., 2020; Green et al., 2019; Sheng, 2025; Briscoe-Tran, 2026).

When submitting a review, employees first provide an overall assessment of the firm on a five-star scale, where higher values indicate greater job satisfaction. In addition to the overall rating, reviewers may evaluate specific workplace dimensions, including work–life balance, culture and values, career opportunities, and compensation and benefits. Glassdoor introduced an additional rating for diversity and inclusion in 2020. Each dimension is rated on a one- to five-star scale, providing structured and comparable measures of employee perceptions across firms and over time.

While these numerical ratings offer standardized indicators of employee sentiment, they are inherently coarse and do not map cleanly to specific dimensions of corporate culture. In particular, broad categories such as "culture and values" are conceptually ambiguous: a high or low score may reflect a wide range of considerations, including managerial quality, ethical standards, workplace flexibility, compensation fairness, or interpersonal climate. As a result, psychological safety—if present—cannot be directly observed in these ratings and cannot be disentangled from other overlapping aspects of the employee experience.

Crucially, alongside the numerical ratings, employees are required to provide textual feedback in the form of short free-text responses describing the positive and negative aspects of working at the firm, subject to a minimum word requirement. These narratives allow employees to articulate specific workplace dynamics in their own words and capture nuanced information that is not fully reflected in categorical ratings, such as leadership behavior, communication norms, and interpersonal interactions. We use these textual responses to construct our measure of psychological safety (PS).

One common concern when using online employee reviews is the potential for selection bias, as individuals with particularly strong positive or negative experiences may be more likely to submit reviews. However, Glassdoor mitigates this concern through a platform-specific feature known as the "give-to-get" mechanism, which requires users to contribute their own review before accessing additional content. Prior research shows that this mechanism reduces selection bias by encouraging participation from employees with more moderate views and leads to a more balanced set of reviews (Marinescu et al., 2018; Sheng, 2025).

Our sample consists of employee reviews for U.S.-listed firms from 2011 to 2021. To ensure sufficient firm-level information, we restrict the sample to firms that receive at least ten reviews during the sample period. This restriction is consistent with prior literature

and helps reduce noise from sparsely reviewed firms. In total, the sample includes 1,892,127 employee reviews covering 2,282 unique firms, with an average of 829 reviews per firm.

We use these employee review data to construct our measure of psychological safety (described in Section 3). We then merge the resulting firm-level measures with accounting information from Compustat and market capitalization and stock return data from CRSP.

We also obtain factor returns for the time-series portfolio regressions. Fama–French factors are obtained from Ken French's Data Library. Q-factors of Hou et al. (2021) are taken from the Global-q website. Mispricing factors of Stambaugh and Yuan (2017) are taken from Bryan Kelly's website. Organizational capital factor data of Eisfeldt and Papanikolaou (2013) are taken from the GitHub link provided on Eisfeldt's website.

Our patent data, which are available only through 2019, and the five aspects of corporate culture constructed from transcripts of earnings calls (Li et al., 2020) are obtained from the WRDS platform. We also use data on firms' stock of intangible capital, computed by Peters and Taylor (2017), from WRDS. Finally, data on the Best Workplaces list are obtained from Boustanifar and Kang (2022), as provided on Boustanifar's webpage.

Our final sample includes 2,025 unique companies with 14,353 firm-year observations.

3 Measuring psychological safety

This section describes our methodology for constructing a firm-year measure of psychological safety (PS) from textual employee reviews on Glassdoor. Following Edmondson (1999), we define psychological safety as employees' shared belief that they can speak up with ideas, questions, concerns, or mistakes without fear of negative consequences.

A central measurement challenge is that employees rarely describe psychological safety using the terminology of organizational theory. In employee reviews, psychological safety is more likely to appear as a latent workplace condition reflected in concrete experiences, interactions, and managerial practices. In particular, the absence of psychological safety is often expressed indirectly through descriptions of leadership behavior, communication norms, feedback practices, or interpersonal dynamics. For example, employees may not explicitly state that they fear speaking up, but may describe managers as punishing dissent, blaming employees for mistakes, discouraging disagreement, ignoring feedback, or dismissing employee concerns in ways that make voice feel unwelcome or risky.

Accordingly, a text-based measure of psychological safety should not be limited to explicit mentions of "psychological safety," "speaking up," or "fear of retaliation." Such an approach would likely miss many economically meaningful references because employees often describe the underlying conditions that enable or inhibit voice rather than naming the construct itself.

Measuring psychological safety therefore requires contextual interpretation rather than simple keyword matching. Our approach is designed to capture both direct references to employees' willingness to speak up and indirect references to the managerial behaviors, communication practices, and interpersonal norms that shape whether speaking up is perceived as safe.

Our approach follows an LLM-labeled supervised-learning design. We use GPT-4o to generate theory-guided psychological safety annotations for a subset of reviews and then fine-tune a transformer-based classifier to apply the measure to the full corpus. This design is well suited to our setting because psychological safety is often expressed indirectly and requires contextual interpretation at the annotation stage, while large-scale empirical analysis requires a fixed, scalable, and reproducible scoring procedure. The LLM provides the contextual interpretation needed to identify nuanced, theory-relevant language in the annotated sample, while the supervised classifier provides a stable measurement model that can be applied consistently across the full corpus. [7]

The measurement procedure proceeds in three steps. First, we use GPT-4o to label 50,000 randomly selected reviews. Second, we fine-tune a transformer-based classifier on these labels. Third, we apply the trained model to the full corpus of Glassdoor reviews and aggregate review-level scores to the firm-year level. Details of the prompt and implementation are provided in Appendix A; we summarize the main features of each step below.

In the first step, each review is evaluated independently using a structured prompt grounded in the organizational psychology literature. The prompt asks GPT-4o to assess whether the review contains evidence about interpersonal risk-taking and the social-emotional environment relevant for psychological safety. Direct expressions include references to fear, retaliation, punishment, embarrassment, or blame associated with speaking up. Indirect expressions include descriptions of the perceived climate for employee voice, such as whether employees can raise concerns, offer feedback, ask questions, admit mistakes, or disagree with managers without negative consequences.

An important construct-boundary issue is that psychological safety is distinct from collaboration, managerial responsiveness, and voice efficacy. Employees may feel safe to offer feedback even when their suggestions are not adopted, and complaints that management "does not listen" do not necessarily imply a lack of psychological safety. At the same time, our prompt instructs GPT-4o to consider tone, implication, and context, rather than only

[7] Methodologically, our approach is related to weak-supervision frameworks that use imperfect or programmatically generated labeling sources to construct training data (Ratner et al., 2017), to recent work using LLM-generated annotations as training labels for supervised classifiers (Pangakis and Wolken, 2024), and, more broadly, to teacher–student approaches that transfer information from larger models to smaller or more specialized models (Hinton et al., 2015). Unlike classical knowledge distillation, however, our student model is trained on LLM-generated review-level labels rather than on the teacher model's full predictive distribution.

explicit statements. As a result, the LLM annotator may classify some reviews about ignored, dismissed, or unwelcome feedback as relevant to psychological safety when such language appears to describe a broader climate in which employee voice is discouraged, unwelcome, or socially costly. These annotations are then used to fine-tune the supervised classifier. Thus, our measure should be interpreted as a proxy for psychological safety, while recognizing that some implicit cases may overlap with related dimensions such as collaboration, managerial openness, or responsiveness.

GPT-4o assigns each review to a five-point ordinal scale that distinguishes explicit from implicit evidence: -2 for explicit lack of psychological safety, -1 for implicit evidence of low psychological safety, 0 for reviews unrelated to psychological safety, +1 for implicit evidence of high psychological safety, and +2 for explicit evidence of high psychological safety. This scoring structure assigns the strongest classifications to explicit references to fear, punishment, retaliation, blame, or embarrassment, and weaker classifications to indirect descriptions suggesting that speaking up is discouraged, unwelcome, or risky. The same prompt is applied uniformly across all reviews.

In the second step, we use the GPT-generated annotations as training data for supervised learning. Specifically, we fine-tune DeBERTa-v3-base for sequence classification. DeBERTa-v3 is a transformer-based encoder model that extends the BERT architecture by incorporating disentangled attention and improving pretraining through ELECTRA-style replaced-token detection and gradient-disentangled embedding sharing (He et al., 2023). We use DeBERTa-v3 because our task requires fine-grained natural language understanding of employee-written text, including indirect references to voice, openness, retaliation, and tolerance for mistakes. Relative to earlier BERT-based encoders, DeBERTa-v3 provides a stronger general-purpose language representation while remaining sufficiently compact and computationally feasible for large-scale scoring. Fine-tuning serves two purposes. First, it allows the model to learn domain-specific linguistic patterns associated with psychological safety in employee reviews, including both explicit and implicit descriptions of the climate for employee voice. Second, it produces a fixed and replicable scoring procedure that can be applied to the full dataset without repeatedly querying a proprietary LLM, substantially reducing computational costs and enhancing reproducibility.[8]

The fine-tuned model closely replicates the GPT-generated annotations in held-out data. In the test sample, the model achieves an accuracy of 80% and a weighted-average F1 score of 0.80 across the five ordinal categories. This level of performance is meaningful given

[8]Although finance research commonly uses finance-domain variants such as FinBERT, our setting differs because the relevant text consists of employee reviews rather than financial disclosures, and the classification target is psychological safety rather than financial sentiment.

that the task requires the model not only to distinguish reviews related and unrelated to psychological safety, but also to separate positive from negative evidence and implicit from explicit statements. Consistent with the ordinal structure of the labels, the confusion matrix shows that most classification errors occur between adjacent categories, such as assigning a review a score of +1 rather than +2, rather than between substantively opposite categories. Thus, the model preserves the main direction and intensity of the GPT classifications in most cases. These results indicate that the fine-tuned classifier provides a reliable and scalable approximation to the GPT-based annotation procedure, allowing us to apply a fixed classification rule to the full sample while preserving replicability and substantially reducing computational costs.

As an additional validation exercise, we construct an independent human-labeled validation sample. After training the classifier, we manually label 450 new employee reviews using the same five-category psychological-safety scale. These reviews are not used in either the training sample or the GPT-labeled test sample. We then compare the fine-tuned model's predictions with the human annotations. The model achieves an accuracy of 75% and a weighted-average F1 score of 0.79 in this independent sample. The confusion matrix shows that most errors are local to the ordinal scale, with misclassifications occurring primarily between adjacent categories rather than across opposite sides of the psychological-safety distribution. This evidence indicates that the classifier does not simply reproduce GPT-generated labels, but also aligns closely with independent human judgments of psychological safety in employee-written text.

In the final step, we apply the fine-tuned classifier to the full corpus of Glassdoor reviews to generate review-level psychological safety scores. We then aggregate these scores to the firm-year level by averaging across reviews. This aggregation captures employees' collective perceptions of the interpersonal climate while reducing idiosyncratic noise in individual reviews.

3.1 Illustrative examples and construct boundaries

To provide intuition for how our PS maps into employee narratives, we examine representative comments that receive low, high, or neutral PS scores. These examples illustrate that the measure captures language related to voice, feedback, error tolerance, and the perceived interpersonal consequences of speaking up. They also help clarify the construct boundary discussed above. In particular, some comments refer directly to fear, punishment, blame, or retaliation, while others refer more indirectly to whether employee input is welcomed, dismissed, or ignored. We interpret the latter cases as weaker and potentially noisier signals

of psychological safety, rather than as definitive evidence that employees would be punished for speaking up.

Comments receiving low PS scores include both direct references to interpersonal risk and indirect descriptions of a workplace climate in which employee voice appears discouraged or unwelcome. The following representative examples are verbatim excerpts from longer employee comments: *"No way you can speak up if something is wrong"*, *"There is little tolerance for mistakes"*, *"Directive head of organization that does not listen to feedback from team"*, and *"Leaders like for people to follow orders not to think. Reward people that aligned to their directives,"* The most direct examples describe fear of speaking up, intolerance for mistakes, or blame.[9] Other examples, such as those referring to leaders who do not listen to feedback or expect employees to follow orders, are more indirect. They are relevant to our measure because they describe a climate in which employee voice appears unwelcome or discouraged, even though they may also reflect related dimensions such as managerial responsiveness or collaboration.

In contrast, comments receiving high PS scores highlight openness to feedback, encouragement of voice, and tolerance for mistakes. Representative excerpts, quoted verbatim from the comments, include: *"Very open culture around feedback and they really value our input on most decisions. We don't necessarily have the last say but there's definitely room to voice our opinions without fear of retaliation"*, *"Freedom to speak out and share ideas and concerns"* *"Accountability and feedback are encouraged amongst all levels"*, and *"Top leadership cares and listens to the feedback"*. Again, some examples provide direct evidence of psychological safety, such as the ability to voice opinions without fear of retaliation or the freedom to share ideas and concerns. Others provide more indirect evidence, such as statements that feedback is encouraged. These comments are not equivalent to direct evidence of psychological safety in a narrow sense because they do not explicitly state whether employees feel safe sharing feedback. However, in the context of employee reviews, such language can provide useful evidence about whether voice is perceived as welcome, valued, or socially safe.[10]

Taken together, these examples highlight the diversity and context dependence with which psychological safety is expressed in employee narratives. Voice, feedback, and error tolerance

[9]Intolerance for mistakes need not imply low psychological safety in all contexts, since some mistakes may reflect negligence or lack of accountability. In employee reviews, however, such language often concerns imperfect outcomes, learning, or experimentation rather than clear negligence. In our setting, the model assigns low PS scores when this language appears in a broader context suggesting fear of blame, punishment, or interpersonal risk. The classification therefore reflects the surrounding text rather than the phrase in isolation.

[10]This distinction is consistent with the prompt's scale, which separates explicit evidence from implicit evidence and allows direct references to fear, punishment, retaliation, blame, or embarrassment to receive the strongest classifications.

appear through varied and often indirect phrasing: sometimes explicitly through references to fear, retaliation, blame, or speaking up, and sometimes through descriptions of hierarchy, autonomy, managerial openness, or responsiveness to input.[11]

Moreover, we observe that PS is not mechanically tied to employees' overall evaluations of their firms. To illustrate this point, the comments below receive high PS scores despite relatively low overall ratings (below the midpoint of the scale) assigned by the employee to the firm. Representative excerpts, quoted verbatim from the comments, include: *"Freedom to try out new ideas"*, *"You are allowed to explore ideas and make a real difference"*, and *"Innovative company that allows and sometimes encourages mistakes to be made"*. These comments suggest that employees may perceive a psychologically safe environment even when they are dissatisfied with other aspects of the workplace.

We also observe comments with low overall ratings but neutral PS scores that focus primarily on compensation, benefits, or work–life balance rather than interpersonal dynamics, such as: *"Good pay. Great work place flexibility policy. Work life balance could be better"*, *"Work life balance is great. Annual hikes are below average"*, and *"Very innovative and cool products. High turnover for most employees"*.

Conversely, we also observe comments with high overall ratings but neutral PS scores. These comments typically emphasize pay, prestige, or general workplace satisfaction without referencing voice, feedback, or error tolerance, for example: *"Bonuses are pretty good each year. Work–life balance is bad. Projects are generally under-staffed"*, *"Culture is great"*, *"the company innovates at a crazy pace. There is more work to do than time to do it. But everyone around you is working super hard. And doing what you love"*, *"Work life balance and smart people around"*, and *"Cool place to work with fun ppl"*.

Taken together, these examples show that our measure captures psychological safety rather than general employee satisfaction. Positive comments do not automatically translate into high PS scores when comments focus on pay, prestige, innovation, culture, or workplace enthusiasm without referencing voice, feedback, interpersonal risk, or error tolerance. Conversely, even when the employee has rated the company with an overall low rating, it can still receive high PS scores when employees describe positively the environment with freedom to speak up, share ideas, or make mistakes. This pattern is consistent with the measure's design: high PS scores are assigned primarily when comments describe an environment in which employees can express concerns, offer feedback, or take interpersonal risks without fear of negative

[11]This heterogeneity makes it difficult to construct a fixed dictionary that reliably captures both positive and negative manifestations of the construct. Dictionary-based approaches rely on predefined word lists and frequency counts, which do not account for sentence-level context, negation, tone, or narrative framing. As a result, they may miss subtle expressions of interpersonal risk or misclassify statements whose meaning depends on surrounding language.

consequences.

4 Empirical analysis

4.1 What correlates with psychological safety?

In this section, we examine the psychological safety (PS) measure and its correlations with firm characteristics and related constructs. The results in this section are purely descriptive and should not be interpreted causally. Our objective is to document how PS covaries with observable firm attributes and existing measures of corporate culture. Doing so serves two purposes: first, to characterize firms with higher PS, and second, to provide partial validation of the PS measure by assessing whether it correlates with conceptually related cultural dimensions.

Consistent with the literature on corporate culture (Edmans, 2011; Guiso et al., 2015; Li et al., 2020, 2026), our specifications include industry and year fixed effects, allowing us to examine cross-sectional differences between firms with relatively high versus low PS within industry–year cells.[12]

Table 2 reports the relation between PS and key firm characteristics, including firm size, Tobin's Q, return on assets, and intangible capital intensity. The table also includes measures of corporate culture constructed from earnings conference call transcripts by Li et al. (2020) (*INTEGRITY*, *TEAMWORK*, *INNOVATION*, *RESPECT*, and *QUALITY*), as well as an indicator for inclusion in the Best Workplaces list (*BW*). We find no meaningful correlation between firm size and PS. Firms with higher PS tend to have higher growth opportunities, as measured by Tobin's Q. The correlation between PS and profitability is weak and, if anything, negative. PS is also positively correlated with intangible capital intensity, although this relation is not particularly robust.

Turning to corporate culture measures, PS is positively correlated with *TEAMWORK* and *INNOVATION*, two dimensions that are conceptually closely related to psychological safety. In contrast, *INTEGRITY* and *QUALITY* exhibit negative correlations with PS. Finally, firms included in the Best Workplaces list tend to have higher levels of PS. Taken together, these patterns provide face validity for the PS measure: dimensions of corporate culture that are theoretically linked to interpersonal trust, collaboration, and idea generation are positively

[12]The central question in the corporate culture literature—including our study—concerns systematic differences across firms rather than short-run within-firm fluctuations. Specifications with firm fixed effects identify the relation exclusively from within-firm time-series variation, which may reflect transitory changes in employee sentiment and measurement noise in culture proxies. Consistent with prior work, we therefore focus on cross-sectional variation across firms within industry–year cells.

associated with PS, while other dimensions capture distinct aspects of organizational culture.

Despite these correlations, industry and year fixed effects, together with the firm characteristics and culture variables considered, explain only about 5% of the variation in PS. This finding suggests that PS captures a dimension of organizational environment that is largely orthogonal to industry affiliation and not easily explained by existing corporate culture measures.

As discussed in Section 2, Glassdoor requires employees to rate their employer using an overall categorical rating from 1 to 5. In addition, employees may provide ratings for specific workplace dimensions, including culture and values (*CULTURE*), work–life balance (*WORKLIFE*), compensation and benefits (*COMPENSATION*), career opportunities (*CAREER*), and diversity and inclusion (*DIVERSITY*), the latter of which was introduced in 2020. These ratings provide a useful benchmark for assessing whether PS captures information that is related to, but distinct from, existing employee perceptions of workplace quality.

Conceptually, psychological safety should be positively related to employees' overall assessment of the firm and to aspects of corporate culture that reflect interpersonal trust and openness. At the same time, PS need not align closely with dimensions such as compensation or career advancement, which reflect different facets of the employment relationship. Table 3 reports the relation between PS and the various Glassdoor ratings.

Consistent with these expectations, we find a positive correlation between PS and the overall firm rating (*RATING*), as well as with *CULTURE*. PS is also positively correlated with *WORKLIFE*, although this relation is weaker and not robust across specifications. In contrast, *COMPENSATION* is negatively correlated with PS, and we find no meaningful correlation between PS and either *CAREER* or *DIVERSITY*. These patterns highlight that PS captures a distinct dimension of the employee experience that is not reducible to material rewards or formal career prospects.

The lack of correlation between PS and *DIVERSITY* is particularly informative. Recent studies using employee review data from sources such as Great Place to Work and Glassdoor analyze diversity and inclusion as a distinct organizational dimension (Briscoe-Tran, 2023; Edmans et al., 2025). However, diversity and inclusion, as reported by employees, primarily reflect demographic representation and inclusive policies, whereas psychological safety concerns whether employees feel able to speak up, challenge decisions, or admit mistakes without fear of negative consequences. A workplace may be demographically diverse and formally inclusive yet lack psychological safety, or vice versa. The absence of any significant correlation between these measures therefore reinforces the conceptual distinction between psychological safety and diversity-related constructs.

Finally, it is important to note that even after accounting for Glassdoor ratings, as well

as industry and year fixed effects, the majority of the variation in PS remains unexplained. This finding underscores two key points. First, the positive correlations with conceptually related ratings provide supporting evidence for the validity of the PS measure. Second, and equally important, PS contains substantial information that is not captured by existing Glassdoor ratings or standard firm characteristics. As such, PS reflects a nuanced aspect of organizational environment that is largely orthogonal to other commonly used employee-based measures.

4.2 Psychological safety and innovation

A central implication of psychological safety is that it facilitates learning, experimentation, and idea generation within organizations. In environments characterized by high psychological safety, employees are more willing to voice novel ideas, challenge prevailing assumptions, and disclose mistakes or failed experiments without fear of negative repercussions (Edmondson, 1999, 2018). These behaviors are critical for innovation, which inherently involves uncertainty, trial and error, and the risk of failure. Organizational and knowledge-based theories of the firm therefore emphasize psychological safety as a foundational condition for effective knowledge creation and innovative activity, particularly in settings that rely on collaboration, decentralized information, and cumulative learning (Argyris, 1990; Grant, 1996; Edmondson, 2018).

Motivated by this theoretical perspective, we examine whether our firm-level measure of PS predicts future innovation outcomes. We focus on patents granted as a standard and widely used proxy for innovative output. Patents have several advantages in this context. First, they provide an objective and externally validated measure of realized innovation. Second, patent grants reflect realized innovative output rather than contemporaneous inputs or intentions. Third, patent data are available at scale and have been extensively used in the economics, finance, and accounting literatures to study firm-level innovation (Griliches, 1990; Hall, 2001; Hall et al., 2005). While patents do not capture all forms of innovation—such as trade secrets or organizational innovations—they reflect codified technological advances that are sufficiently novel to warrant legal protection, making them particularly suitable for assessing whether organizational attributes translate into tangible innovative output.

Table 4 reports the relation between PS and the logarithm of future patent grants. Across specifications, we find a strong and positive association between PS and subsequent innovation. A one-standard-deviation increase in PS is associated with approximately 4% more patents. This relation is robust to controlling for firm size, market-to-book ratio, and R&D expenditures. As expected, firm size and R&D expenditures are themselves strongly

correlated with future patenting activity. Importantly, however, the estimated effect of PS remains economically and statistically significant after accounting for these traditional determinants of innovation.

The effect of PS also persists—and in fact increases in magnitude—when we further control for the firm's overall Glassdoor rating, as well as alternative measures of corporate culture constructed from earnings conference call transcripts following Li et al. (2020). Moreover, controlling for Glassdoor's culture and values rating does not attenuate the estimated coefficient on PS. These results indicate that PS captures information relevant for innovation that is not subsumed by broader measures of employee satisfaction or by existing proxies for corporate culture (Edmans, 2011; Li et al., 2020; Boustanifar and Kang, 2022).

Interestingly, the Glassdoor ratings themselves do not exhibit a positive relation with future innovation. The overall firm rating is statistically insignificant and, if anything, enters with a negative coefficient. This contrast highlights an important distinction between general employee satisfaction and the specific interpersonal climate that supports risk-taking and idea generation. While high satisfaction may reflect compensation, amenities, or work–life balance, these features do not necessarily foster the open communication, candid feedback, and tolerance for failure that innovation requires (Edmondson, 2018). We likewise find no relation between Glassdoor's culture and values rating and future patenting activity, suggesting that broad measures of corporate culture—which aggregate multiple organizational dimensions—are not, by themselves, associated with innovative output. Among the culture variables from Li et al. (2020), *TEAMWORK* is the only measure that exhibits a marginal association with future patents, consistent with the view that collaboration-related dimensions are most closely linked to innovation.

The results are robust to the inclusion of industry-by-year fixed effects, as shown in Appendix Table A1, and to estimating Poisson regressions instead of OLS to address concerns about applying logarithmic transformations to count data (Appendix Table A2).

We also examine citation-based measures of innovation to assess whether PS predicts not only the quantity of innovation but also its ex post technological impact. We do not find a statistically significant relation between PS and citation-based measures; these results are reported in Table A3. This pattern suggests that psychological safety primarily affects the generation and disclosure of innovative ideas—reflected in a higher number of patents—rather than systematically influencing the downstream technological influence or diffusion of those innovations. Citation counts depend on a variety of factors beyond organizational climate, including technological trajectories, field-specific citation practices, strategic patenting behavior, and external adoption, which are largely orthogonal to workplace psychological safety. Overall, these results suggest that psychological safety is more closely

related to the quantity of innovation than to its ex post technological impact.

Taken together, these findings serve two complementary purposes. First, the fact that PS predicts future patents—even after controlling for firm characteristics, related employee ratings, alternative culture measures, and industry–year fixed effects—provides external validation that our measure captures a theoretically meaningful and economically relevant organizational attribute. Second, the results contribute to the innovation literature by identifying psychological safety as a firm-level condition that facilitates innovative output. Unlike traditional innovation inputs such as R&D spending, psychological safety appears to be associated with the efficiency with which organizations convert human capital and knowledge into novel ideas, consistent with theories emphasizing the role of organizational climate in shaping learning, coordination, and experimentation (Edmondson, 2018).

At the same time, we acknowledge a common caveat that applies to this analysis, as well as to much of the related literature using patent-based outcomes. Specifically, these relationships should not be interpreted causally. Even with extensive controls, it is not possible to fully account for all unobserved firm capabilities—such as managerial quality, organizational design, or inherent innovative capacity—that may be correlated with both higher psychological safety and superior subsequent real outcomes. Nevertheless, the documented associations remain informative as external validation of our measure and are consistent with theoretical predictions linking psychological safety to learning and experimentation.

More broadly, accounting-based outcomes are also inherently subject to similar endogeneity concerns, as they reflect realized firm performance that is jointly determined by a wide range of observable and unobservable factors. As a result, such tests are ill suited for drawing causal inferences about the real effects of psychological safety, even when incorporating rich sets of controls and fixed effects. For this reason, our focus shifts in the next sections to market valuation and stock returns, which provide a fundamentally different empirical setting. Under standard asset-pricing logic, stock prices should incorporate all publicly available information about firms' organizational quality and future prospects. To the extent that psychological safety merely reflects underlying firm quality, this information should already be reflected in prices. Examining whether psychological safety predicts future returns therefore allows us to assess the joint hypotheses that psychological safety is financially material to the firm and that it is not fully incorporated into market valuations.

4.3 Psychological safety and current market valuation

In this section, we examine whether the value of psychological safety is reflected in firms' current market valuations. If psychological safety is value-relevant and recognized by the

market, firms with higher psychological safety should exhibit higher valuation ratios.

Table 5 reports the relation between PS and the market-to-book ratio. Across specifications, we control for the firm's overall Glassdoor rating, size, leverage, and intangible capital, and we consider alternative fixed-effect structures, including specifications with no fixed effects, year fixed effects, and both year and industry fixed effects. Across all specifications, we find no evidence that firms with higher PS command higher market valuations. In contrast, firms with higher Glassdoor ratings (reflecting overall employee satisfaction), larger firms, firms with greater leverage, and firms with higher intangible capital exhibit higher market-to-book ratios. Table A4 reports similar results for another commonly used valuation metric, the price-to-earnings ratio, showing that firms with higher PS do not trade at higher current valuations.

These findings suggest that the market does not appear to incorporate the potential long-term benefits associated with psychological safety into contemporaneous valuations. If psychological safety is indeed financially relevant but not fully reflected in prices, one would expect firms with higher psychological safety to earn higher future returns. We investigate this possibility in the next sections.

4.4 Psychological safety and risk-adjusted returns

4.4.1 Long–short portfolios

We begin by examining whether PS, our measure of psychological safety, predicts future stock returns using standard equal-weighted portfolio sorts. In each year, we sort firms into quintiles based on their PS score, with Quintile 1 containing firms with the lowest PS scores and Quintile 5 containing firms with the highest scores. To mitigate timing concerns, portfolios are formed after the measurement year and held from July of year $(t+1)$ through June of year $(t+2)$. Monthly portfolio returns are computed as the equal-weighted average return of all firms in each PS quintile. We then construct a long–short portfolio that buys firms in the highest PS quintile and shorts firms in the lowest quintile.

To estimate risk-adjusted returns, we conduct monthly time-series regressions of the long–short portfolio returns using three commonly used factor models. Specifically, we consider the Fama–French five-factor model augmented with momentum (Fama and French, 2015) (FF6), the q-factor model of Hou et al. (2021) (Q5), and the mispricing four-factor model of Stambaugh and Yuan (2017) (SY4). We multiply monthly portfolio returns and monthly factor returns by 12 before estimating the factor regressions. This scaling expresses intercepts as annualized alphas while leaving factor loadings unchanged. Statistical inference is based on Newey–West standard errors to account for heteroskedasticity and autocorrelation

in monthly returns.

Table 6 reports the results. Each column corresponds to a different factor model. Across all factor specifications, we find positive and statistically significant risk-adjusted returns associated with PS. The estimated annualized alphas range from 4.57% (t-stat = 2.90) under the Q5 model to 5.99% (t-stat = 3.34) under the SY4 model. The FF6 model yields an alpha of 5.18% (t-stat = 3.12). These results indicate that firms with high PS earn significantly higher future returns than firms with low PS, even after controlling for standard factor exposures.

4.4.2 Sensitivity to weighting method

Table 6 reports results based on standard equal-weighted portfolios. A potential concern is that these results could be disproportionately influenced by small firms. To assess whether the baseline findings are sensitive to portfolio weighting choices, we next examine the performance of PS portfolios under alternative weighting schemes. Specifically, we construct size-neutral equal-weighted and value-weighted portfolios. Each year, we independently sort firms into PS quintiles and size terciles, where size is measured using beginning-of-period June market equity. Within each PS quintile, returns are first computed separately within each size tercile. We then average these returns equally across the three size groups, ensuring that the resulting PS portfolios are not driven by differences in size composition. Panel A reports results using equal-weighted returns within each size-by-PS cell, while Panel B reports results using value-weighted returns within each cell.

The results in Table 7 show that the estimated alphas remain statistically significant and economically sizable under both alternative weighting methods. In Panel A, the alphas for size-neutral equal-weighted portfolios range from 3.03% to 4.53%. In Panel B, the alphas for size-neutral value-weighted portfolios range from 3.85% to 5.19%. These findings indicate that the outperformance associated with psychological safety is not concentrated among small firms or driven by a particular portfolio-weighting convention. Rather, the evidence is robust to explicitly balancing portfolios across size groups and to using value weights within size-by-PS cells. Given this robustness, in the remainder of the paper we report results based on the baseline equal-weighted portfolio construction, while noting that the main findings are similar under alternative weighting methods.

4.4.3 Long- and short-leg returns

The previous sections show that long–short portfolios formed on PS earn positive and statistically significant risk-adjusted returns. This return differential may arise because firms

with high PS earn unusually high returns, because firms with low PS earn unusually low returns, or both. To examine which side of the portfolio drives the results, we estimate factor regressions separately for the high- and low-PS legs, using the same factor models as in Table 6.

Table 8 reports the results across the three factor models. The evidence indicates that the abnormal return associated with PS is driven primarily by the high-PS leg. Firms in the highest PS quintile earn positive and statistically significant alphas, whereas firms in the lowest PS quintile have alphas that are small in magnitude and statistically insignificant. For example, under the FF6 model, the high-PS leg earns an annualized alpha of 4.80% (t-stat = 2.66), compared with an alpha of −0.38% (t-stat = −0.32) for the low-PS leg.

From an economic perspective, this asymmetry suggests that the market primarily underprices the long-term benefits associated with high psychological safety, rather than overpricing firms with low psychological safety. Psychological safety reflects intangible organizational attributes that may not be readily observable or whose value may be difficult for investors to assess in real time, leading to gradual incorporation into prices. In contrast, firms with low psychological safety may not be systematically overvalued, or their weaknesses may be more readily observable, resulting in weaker and less consistent underperformance.

4.4.4 Glassdoor ratings and abnormal returns

Although Table 3 shows that most of the variation in PS is not explained by Glassdoor ratings, a natural concern is whether the PS premium documented above reflects the effects of Glassdoor ratings, which capture broader employee satisfaction.

Before addressing this concern directly, it is useful to note that the results presented in previous sections do not suggest that Glassdoor ratings, $RATING$, capture the same economic content as PS, our measure of psychological safety. For example, Table 4 shows that PS is positively related to future patents, whereas the relation between $RATING$ and patents is negative and statistically insignificant. Similarly, Table 5 shows that firms with higher $RATING$ have higher current valuations, whereas PS is not significantly related to current valuations. This evidence suggests that, consistent with our argument, Glassdoor ratings and PS capture distinct aspects of firms' organizational environments with potentially different economic implications.

To address this concern more directly, we conduct two analyses. First, we examine the returns to Glassdoor ratings by forming portfolios based on firms' overall Glassdoor ratings instead of PS. We then estimate time-series factor regressions analogous to those reported in Table 6. The results, reported in Panel A of Table 9, show no evidence that long–short portfolios sorted on Glassdoor ratings earn positive or statistically significant abnormal

returns. The estimated alphas across factor models are economically small, ranging from −16 to 64 basis points per year depending on the model, and are statistically indistinguishable from zero, with t-statistics below 0.40.[13]

In the second analysis, we residualize PS with respect to Glassdoor ratings by taking the residual from a regression of PS on Glassdoor ratings. We then form portfolios based on this residualized measure, which is orthogonal by construction to Glassdoor ratings. The estimated alphas for these portfolios are reported in Panel B of Table 9. The alphas are positive and statistically significant across all factor models. In terms of magnitude, the alphas are very similar to those obtained without residualizing PS; in two models, FF6 and Q5, the alphas for residualized PS are even slightly larger.

These findings suggest that the abnormal returns associated with PS are not driven by Glassdoor ratings. Importantly, the absence of abnormal returns associated with Glassdoor ratings does not imply that overall employee satisfaction is unimportant. Rather, it suggests that broad and highly visible measures of employee satisfaction, such as Glassdoor ratings, may already be reflected in market prices. Consistent with this interpretation, Table 5 shows that firms with higher Glassdoor ratings trade at significantly higher contemporaneous valuation ratios. In contrast, psychological safety captures a more nuanced and less observable organizational attribute whose economic relevance may not be fully incorporated into current valuations and may therefore be associated with higher subsequent returns.

4.4.5 Exposure to intangible factors

Another concern is that PS may proxy for other dimensions of intangible capital that are already known to be related to stock returns. In particular, firms with high PS may systematically invest more in research and development or organizational capital, both of which prior studies have shown to be related to stock returns (Chan et al., 2001; Eisfeldt and Papanikolaou, 2013).[14] To assess whether the abnormal returns associated with PS simply reflect exposure to these established intangible-related factors, we examine the robustness of our results after controlling for factor returns constructed from R&D intensity and organizational capital.

Table 10 reports the results from the FF6 model augmented with the R&D factor and the organizational capital factor, included either separately in Columns 2–3 or jointly in Column 4. For reference, Column 1 reports the baseline FF6 specification. The results show that the

[13]These results are consistent with Green et al. (2019), who find no evidence that firms with higher Glassdoor ratings earn excess returns.

[14]The proposed mechanisms linking intangible capital to stock returns differ across the literature: Chan et al. (2001) emphasize the underpricing of R&D, whereas Eisfeldt and Papanikolaou (2013) interpret the organizational capital premium as compensation for risk.

alpha of the PS portfolio remains positive and statistically significant across all specifications. If anything, the magnitude of the alpha increases, from 5.18% (t-stat = 3.12) in the baseline FF6 model to 5.50% (t-stat = 3.11) when both intangible-related factors are included.

Overall, these findings indicate that the abnormal returns associated with PS are not explained by exposure to R&D intensity or organizational capital factors. Instead, PS appears to capture a distinct dimension of firm-level intangible characteristics that is not subsumed by existing intangible-related risk factors.

4.5 Earnings surprises and market reactions

In the previous section, we show that firms with higher PS earn positive risk-adjusted returns after controlling for a broad set of common risk factors. These findings are consistent with the possibility that psychological safety is not fully reflected in contemporaneous market valuations, as we also document that firms with higher PS do not trade at higher valuation ratios. If psychological safety is indeed economically relevant but underappreciated by the market, its value should be incorporated gradually over time. One natural mechanism through which this incorporation can occur is via earnings announcements.

Specifically, if market participants and analysts underweight psychological safety when forming expectations because it is not readily observable, firms with higher psychological safety should systematically outperform earnings expectations, resulting in positive earnings surprises. These surprises should, in turn, be accompanied by positive stock price reactions around earnings announcements, contributing to the excess returns documented earlier.

We test this implication using quarterly earnings announcements. Earnings surprises are measured using standardized unexpected earnings based on the median analyst forecast (SUE), and market reactions are measured using cumulative abnormal returns over a three-day window surrounding the earnings announcement (CAR). Following the literature (Edmans, 2011; Atilgan et al., 2023), we include standard firm-level controls, including the natural logarithm of lagged quarterly market capitalization ($SIZE$) and the natural logarithm of the lagged quarterly book-to-market ratio (BM). We estimate specifications with alternative sets of fixed effects, including year, industry, year and industry, and year×industry fixed effects.

The results are reported in Table 11. Columns 1–4 show that, across all specifications, firms with higher PS experience significantly higher quarterly earnings surprises. Columns 5–8 show that these firms also earn significantly higher abnormal returns around earnings announcements. Together, these results indicate that firms with higher PS systematically outperform market expectations and that this information is incorporated into prices at earnings announcements.

Overall, these findings are consistent with a gradual incorporation of the value of psychological safety into market prices. Rather than being reflected in contemporaneous valuation ratios, the economic relevance of psychological safety appears to materialize through future earnings surprises and corresponding market reactions, providing a plausible mechanism for the excess returns documented in earlier sections.

4.6 Cross-sectional stock returns

The portfolio-level evidence presented above indicates that firms with higher psychological safety (PS) earn significantly higher long–short excess returns, even after controlling for standard risk factors and factors related to intangible assets. Portfolio sorts play a central role in the asset-pricing literature because they deliver economically interpretable return spreads, are relatively robust to measurement error in firm characteristics, and do not impose linearity assumptions on the pricing relation. These features are particularly important for organizational attributes such as psychological safety, which are inherently noisy and difficult to measure precisely.

At the same time, portfolio sorts are less well suited to isolating conditional return predictability in the presence of correlated firm characteristics. We therefore complement the portfolio analysis with Fama–MacBeth regressions, which allow us to control flexibly for a wide range of firm-level variables and to assess whether PS carries incremental return-predictive information beyond established predictors. Importantly, Fama–MacBeth regressions address a different question than portfolio sorts. While portfolio sorts capture economically meaningful return differentials that may reflect bundled or nonlinear effects, Fama–MacBeth regressions estimate the average conditional relation between a characteristic and future returns, controlling for other firm characteristics. When the economic relevance of a characteristic is heterogeneous across firms or concentrated in specific environments, the average Fama–MacBeth coefficient may therefore be weak or statistically insignificant even when portfolio return spreads are sizable.

This consideration is particularly relevant for psychological safety, whose effects are likely to depend on firms' innovation intensity and labor composition. Moreover, because PS is measured with noise, especially when aggregated across different types of workers, measurement error can further attenuate average regression coefficients. Accordingly, we focus not only on the average pricing effect of PS, but also on interaction terms and decompositions that allow its return-predictive implications to vary across firms and worker types.

The firm characteristics include beta, the logarithm of market capitalization, the logarithm of the book-to-market ratio, 12-month momentum, profitability, asset growth, trading volume,

return skewness, the logarithm of stock price, leverage, net equity issuance, illiquidity, and intangible capital intensity from Peters and Taylor (2017). We also control for firms' overall Glassdoor ratings and an indicator for inclusion in the Best Workplaces list. Statistical inference is based on Newey–West standard errors.

Table 12 reports the results. Columns 1 and 2 examine the average effect of PS. Column 1 controls for standard firm characteristics, while column 2 additionally includes Glassdoor ratings and the Best Workplaces indicator. In both specifications, the estimated coefficient on PS is positive but statistically insignificant. This result suggests that the return predictability of PS may not be uniform across firms, motivating our analysis of heterogeneity below.

Organizational theory predicts that psychological safety should be particularly relevant in firms that rely more heavily on innovation, as it facilitates idea generation, experimentation, and learning from failure. Motivated by this insight, we interact PS with firm-level R&D intensity, measured as R&D scaled by market capitalization. The interaction term is positive and statistically significant in column 3 of Table 12, indicating that the return premium associated with PS is concentrated among more innovation-intensive firms. Notably, the coefficient on R&D intensity itself is not statistically significant.

We further explore heterogeneity across labor types by decomposing PS into separate measures based on reviews from white-collar and blue-collar employees. This decomposition reflects the idea that psychological safety should be most consequential in environments where knowledge sharing, discretionary effort, and creative problem solving are central. As shown in column 4 of Table 12, PS constructed from white-collar employee reviews is positively and significantly associated with future stock returns, whereas the corresponding measure based on blue-collar reviews exhibits no relation with returns. In column 5, we further find that the return predictability associated with white-collar PS is particularly strong among firms with higher R&D intensity. These findings indicate that the pricing implications of psychological safety are concentrated in the component of the measure derived from white-collar employees, whose roles are more directly tied to innovation and strategic decision-making.

Overall, the cross-sectional results suggest that psychological safety is most value-relevant, and most likely to be underappreciated, when firms operate in environments where knowledge creation and experimentation are central to the production process (Edmondson, 1999, 2018). In such settings, psychological safety facilitates the expression of novel ideas, the communication of negative information, and learning from failure, and is associated with higher subsequent stock returns.

5 Conclusion

In an influential survey of executives, Graham et al. (2022) report that 92% of respondents believe improving corporate culture would increase firm value. In their comprehensive review, Gorton et al. (2023) conclude that "much remains to be studied and developed on corporate culture," emphasizing that the appropriate empirical approach is to "unpack it." Recent research has followed this agenda by isolating specific dimensions of corporate culture, including workplace flexibility (Au et al., 2021), sexual harassment (Au et al., 2024), access to meaningful projects (Tambe et al., 2020), diversity, equity, and inclusion (Cai et al., 2022; Corritore et al., 2020; Briscoe-Tran, 2023; Edmans et al., 2025), and stakeholder-specific perspectives on corporate culture (Li et al., 2026).

This paper examines the measurement and economic implications of psychological safety at work. Using textual employee reviews from Glassdoor, we construct a novel firm-level measure of psychological safety (PS) that captures employees' perceptions of voice, feedback, and interpersonal risk. By combining LLM-based annotation with supervised machine learning, we develop a scalable and replicable measure that is empirically distinct from standard employee ratings and broader measures of corporate culture.

We document several key findings. PS predicts future innovation, but is not associated with higher contemporaneous market valuations. At the same time, firms with higher PS earn significant risk-adjusted stock returns. These returns are robust across multiple factor models and portfolio constructions and are strongest among innovation-intensive firms and when PS is measured using white-collar employee reviews. We further show that firms with higher PS experience systematically positive earnings surprises and stronger market reactions around earnings announcements, suggesting that the economic value of psychological safety is incorporated into prices gradually over time.

Our paper points to several promising avenues for future research. Although our evidence is consistent with the organizational behavior literature's emphasis on psychological safety as a foundation for learning and innovation, psychological safety may affect firm outcomes through a broader set of channels. By encouraging employee voice, it may not only facilitate the generation of new ideas but also help organizations identify and terminate bad ideas more quickly, thereby improving capital allocation and operational efficiency. Psychological safety may also strengthen internal monitoring, reduce the hoarding of bad news, and enable the earlier detection of fraud, compliance violations, and operational risks, potentially lowering crash risk and other forms of downside risk. In addition, it may encourage employees to question established practices and help firms adapt more effectively to disruptive forces and adverse shocks. Future research could examine these channels directly and investigate the

organizational conditions under which they are most important. In particular, the effects of psychological safety are likely to vary with firms' technologies, workforce composition, and stages of the corporate life cycle. We hope that the measure introduced in this paper will facilitate research on these questions and contribute to a more detailed understanding of when and how psychological safety creates firm value.

References

Aguda, T. D., Siddagangappa, S., Kochkina, E., Kaur, S., Wang, D., and Smiley, C. (2024). Large language models as financial data annotators: A study on effectiveness and efficiency. In Calzolari, N., Kan, M.-Y., Hoste, V., Lenci, A., Sakti, S., and Xue, N., editors, *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 10124–10145, Torino, Italia. ELRA and ICCL.

Argyris, C. (1990). *Overcoming Organizational Defenses: Facilitating Organizational Learning*. Allyn and Bacon, Boston.

Asirvatham, H., Mokski, E., and Shleifer, A. (2026). GPT as a measurement tool. Working Paper 34834, National Bureau of Economic Research.

Atilgan, Y., Demirtas, K. O., Edmans, A., and Gunaydin, A. D. (2023). Does the carbon premium reflect risk or mispricing? Unpublished working paper, Sabanci University, Sabanci University, London Business School, Sabanci University.

Au, S., Dong, M., and Tremblay, A. (2024). How much does workplace sexual harassment hurt firm value? *Journal of Business Ethics*, 190:861–883.

Au, S.-Y., Dong, M., and Tremblay, A. (2021). Employee flexibility, exogenous risk, and firm value. *Journal of Financial and Quantitative Analysis*, 56(3):853–884.

Bams, D., Renders, A., Sheykhi, S., and Struyfs, K. (2025). Employee satisfaction and firm value: The importance of strategic context and color of the collar. *Available at SSRN 5692024*.

Barth, A., Mansouri, S., and Wöbbeking, F. (2023). "Let me get back to you" — a machine learning approach to measuring nonanswers. *Management Science*, 69(10):6333–6348.

Bodnaruk, A., Loughran, T., and McDonald, B. (2015). Using 10-K text to gauge financial constraints. *Journal of Financial and Quantitative Analysis*, 50(4):623–646.

Boustanifar, H. and Kang, Y. D. (2022). Employee satisfaction and long-run stock returns, 1984-2020. *Financial Analysts Journal*, 78(3):129–151.

Boustanifar, H. and Kang, Y. D. (2024). The brand premium. *The Review of Financial Studies*, 38(1):294–336.

Boustanifar, H. and Mansouri, S. (2026). Management disclosure and media coverage. *Journal of Financial and Quantitative Analysis*. Forthcoming.

Boustanifar, H. and Verriest, A. (2026). Intangible liabilities. *Management Science*. Forthcoming.

Briscoe-Tran, H. (2023). Are firms sacrificing flexibility for diversity and inclusion? SSRN Working Paper.

Briscoe-Tran, H. (2026). Do employees have useful information about firms' ESG practices? *Management Science*. Forthcoming.

Cai, W., Dey, A., Grennan, J., Pacelli, J., and Qiu, L. (2022). Do diverse directors influence dei outcomes? Technical Report SSRN 4232434, SSRN.

Campbell, D. W. and Shang, R. (2022). Tone at the bottom: Measuring corporate misconduct risk from the text of employee reviews. *Management Science*, 68(9):7034–7053.

Chan, L. K. C., Lakonishok, J., and Sougiannis, T. (2001). The stock market valuation of research and development expenditures. *Journal of Finance*, 56(6):2431–2456.

Choi, G.-Y. and Kim, A. G. (2025). Firm-level tax audits: A generative AI-based measurement. Working Paper 4645865, SSRN.

Corritore, M., Goldberg, A., and Srivastava, S. B. (2020). Duality in diversity: How intrapersonal and interpersonal cultural heterogeneity relate to firm performance. *Administrative Science Quarterly*, 65(2):359–394.

Dube, S. and Zhu, C. (2021). The disciplinary effect of social media: Evidence from firms' responses to glassdoor reviews. *Journal of Accounting Research*, 59(5):1783–1825.

Duhigg, C. (2016). What google learned from its quest to build the perfect team. *The New York Times Magazine*.

Edmans, A. (2011). Does the stock market fully value intangibles? employee satisfaction and equity prices. *Journal of Financial Economics*, 101(3):621–640.

Edmans, A., Flammer, C., and Glossner, S. (2025). (diversity) equity and inclusion. SSRN Working Paper.

Edmans, A., Pu, D., Zhang, C., and Li, L. (2024). Employee satisfaction, labor market flexibility, and stock returns around the world. *Management Science*, 70(7):4357–4380.

Edmondson, A. (1999). Psychological safety and learning behavior in work teams. *Administrative science quarterly*, 44(2):350–383.

Edmondson, A. C. (2018). *The Fearless Organization: Creating Psychological Safety in the Workplace for Learning, Innovation, and Growth*. John Wiley & Sons, Hoboken, NJ.

Edmondson, A. C. and Lei, Z. (2014). Psychological safety: The history, renaissance, and future of an interpersonal construct. *Annual Review of Organizational Psychology and Organizational Behavior*, 1(1):23–43.

Eisfeldt, A. L. and Papanikolaou, D. (2013). Organization capital and the cross-section of expected returns. *The Journal of Finance*, 68(4):1365–1406.

Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

Filipović, Z. M. and Wagner, A. F. (2026). The intangibles song in takeover announcements: Good tempo, hollow tune. *The Review of Financial Studies*. Forthcoming.

Giglio, S., Kuchler, T., Stroebel, J., and Zeng, X. (2025). Biodiversity risk. *Review of Finance*, page *forthcoming*.

Gilardi, F., Alizadeh, M., and Kubli, M. (2023). ChatGPT outperforms crowd workers for text-annotation tasks. *Proceedings of the National Academy of Sciences*, 120(30).

Gorton, G. B., Grennan, J., and Zentefis, A. K. (2023). Corporate culture. *Annual Review of Financial Economics*, 15:1–26.

Graham, J. R., Grennan, J., Harvey, C. R., and Rajgopal, S. (2020). Corporate culture: The interview evidence. *Journal of Applied Corporate Finance*, 32(1):60–73.

Graham, J. R., Grennan, J., Harvey, C. R., and Rajgopal, S. (2022). Corporate culture: Evidence from the field. *Journal of Financial Economics*, 146(2):552–593.

Grant, R. M. (1996). Toward a knowledge-based theory of the firm. *Strategic Management Journal*, 17(S2):109–122.

Green, T. C., Huang, R., Wen, Q., and Zhou, D. (2019). Crowdsourced employer reviews and stock returns. *Journal of Financial Economics*, 134(1):236–251.

Grennan, J. and Li, K. (2023). Corporate culture: A review and directions for future research. In Hilary, G. and McLean, R. D., editors, *Handbook of Financial Decision Making*, pages 112–132. Edward Elgar Publishing.

Griliches, Z. (1990). Patent statistics as economic indicators: A survey. *Journal of Economic Literature*, 28(4):1661–1707.

Guiso, L., Sapienza, P., and Zingales, L. (2015). The value of corporate culture. *Journal of Financial Economics*, 117(1):60–76.

Hall, B. H. (2001). The stock market's valuation of r&d investment during the 1980s. *American Economic Review Papers and Proceedings*, 91(2):48–52.

Hall, B. H., Jaffe, A. B., and Trajtenberg, M. (2005). Market value and patent citations. *RAND Journal of Economics*, 36(1):16–38.

He, P., Gao, J., and Chen, W. (2023). DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. *arXiv preprint arXiv:2111.09543*.

Hinton, G., Vinyals, O., and Dean, J. (2015). Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*.

Hoberg, G. and Phillips, G. (2025). Scope, scale, and concentration: The 21st-century firm. *Journal of Finance*, 80(1):415–466.

Hou, K., Mo, H., Xue, C., and Zhang, L. (2021). An augmented q-factor model with expected growth. *Review of Finance*, 25(1):1–41.

Huang, K., Li, M., and Markov, S. (2020). What do employees know? evidence from a social media platform. *The Accounting Review*, 95(2):199–226.

Huang, M., Li, P., Meschke, F., and Guthrie, J. P. (2015). Family firms, employee satisfaction, and corporate performance. *Journal of Corporate Finance*, 34:108–127.

Jha, M., Qian, J., Weber, M., and Yang, B. (2024). ChatGPT and corporate policies. NBER Working Paper 32161, National Bureau of Economic Research.

Kalmenovitz, J. (2023). Regulatory intensity and firm-specific exposure. *Review of Financial Studies*, 36(8):3311–3347.

Lee, Y., Ng, S., Shevlin, T., and Venkat, A. (2021). The effects of tax avoidance news on employee perceptions of managers and firms: Evidence from glassdoor. com ratings. *The Accounting Review*, 96(3):343–372.

Li, E. X., Shen, M., Tu, Z., and Zhou, D. (2025). The promise and peril of generative AI: Evidence from GPT as sell-side analysts. *SSRN Electronic Journal*. SSRN Working Paper Series, Posted June 25, 2023; Last revised December 30, 2025.

Li, K., Mai, F., Shen, R., and Yan, X. (2020). Measuring corporate culture using machine learning. *The Review of Financial Studies*, 34(7):3265–3315.

Li, K., Mai, F., Shen, R., Yang, C., and Zhang, T. (2026). Dissecting corporate culture using generative ai: Insights from analyst reports. *Review of Financial Studies*, 39(1):253–296.

Lopez-Lira, A. and Tang, Y. (2026). Can chatgpt forecast stock price movements? return predictability and large language models. *Journal of Financial Economics*. Forthcoming.

Marinescu, I., Klein, N., Chamberlain, A., and Smart, M. (2018). Incentives can reduce bias in online reviews. Technical report, National Bureau of Economic Research.

Pangakis, N. and Wolken, S. (2024). Knowledge distillation in automated annotation: Supervised text classification with LLM-generated training labels. In Card, D., Field, A., Hovy, D., and Keith, K., editors, *Proceedings of the Sixth Workshop on Natural Language Processing and Computational Social Science (NLP+CSS 2024)*, pages 113–131, Mexico City, Mexico. Association for Computational Linguistics.

Peters, R. H. and Taylor, L. A. (2017). Intangible capital and the investment-q relation. *Journal of Financial Economics*, 123(2):251–272.

Ratner, A., Bach, S. H., Ehrenberg, H., Fries, J., Wu, S., and Selsam, D. (2017). Snorkel: Rapid training data creation with weak supervision. In *Proceedings of the VLDB Endowment*, volume 11, pages 269–282.

Sautner, Z., Lent, L. V., Vilkov, G., and Zhang, R. (2023). Firm-level climate change exposure. *Journal of Finance*, 78(3):1449–1498.

Sheng, J. (2025). Asset pricing in the information age: Employee expectations and stock returns. *Review of Asset Pricing Studies*, 15(1):74–101.

Stambaugh, R. F. and Yuan, Y. (2017). Mispricing factors. *Review of Financial Studies*, 30(4):1270–1315.

Tambe, P., Ye, X., and Cappelli, P. (2020). Paying to program? engineering brand and high-tech wages. *Management Science*, 66(11):5149–5169.

Törnberg, P. (2023). ChatGPT-4 outperforms experts and crowd workers in annotating political twitter messages with zero-shot learning.

Vaughan, D. (1996). *The Challenger Launch Decision: Risky Technology, Culture, and Deviance at NASA*. University of Chicago Press.

Table 1: Descriptive Statistics

This table shows the descriptive statistics of the main variables used in the paper. *PS* is the measure of firm-level psychological safety constructed from the employee comments in Glassdoor.*RATING* is the firm's overall employee rating on Glassdoor. *COMPENSATION*, *WORKLIFE*, *DIVERSITY*, *CAREER*, and *CULTURE* are employee ratings for compensation and benefits, work–life balance, diversity and inclusion, career opportunities, and culture and values, respectively. *SIZE* is the logarithm of market capitalization. *MB* is the logarithm of market-to-book ratio. *TOBINQ* is Tobin's Q. *ROA* is return on assets. *LEVERAGE* is total debt to assets. *INT_CAP* is the ratio of capitalized intangible capital to assets, following (Peters and Taylor, 2017). *BW* is an indicator for firms included in the Best Workplaces list compiled by Great Places to Work in America, obtained from Boustanifar and Kang (2022). *INTEGRITY*, *TEAMWORK*, *INNOVATION*, *RESPECT*, and *QUALITY* are measures of corporate culture from (Li et al., 2020).

	count	mean	sd	p25	p50	p75
PS	14,353	0.02	0.42	-0.14	0.00	0.19
RATING	14,353	3.30	0.75	2.94	3.33	3.76
COMPENSATION	14,353	2.93	0.81	2.50	3.00	3.40
WORKLIFE	14,353	2.90	0.83	2.46	2.90	3.34
DIVERSITY	3,377	2.71	0.92	2.32	2.75	3.16
CAREER	14,353	2.68	0.80	2.28	2.69	3.05
CULTURE	13,504	2.80	0.88	2.33	2.79	3.27
SIZE	14,353	7.43	2.15	6.08	7.53	8.88
MB	13,498	0.91	0.92	0.26	0.80	1.44
ROA	12,215	-0.04	0.32	-0.03	0.04	0.08
LEVERAGE	14,190	0.26	0.24	0.07	0.23	0.39
TOBINQ	14,190	1.95	2.10	0.81	1.29	2.28
INT_CAP	14,353	0.64	0.68	0.18	0.56	0.84
BW	14,353	0.02	0.13	0.00	0.00	0.00
INTEGRITY	11,861	2.33	1.16	1.54	2.11	2.85
TEAMWORK	11,861	2.48	1.63	1.36	2.02	3.11
INNOVATION	11,861	5.55	3.33	3.23	4.75	6.95
RESPECT	11,861	3.23	2.31	1.63	2.63	4.14
QUALITY	11,861	2.54	1.63	1.40	2.13	3.23

Table 2: What correlates with psychological safety?

This table examines which firm fundamentals and cultural variables are correlated with our measure of psychological safety, *PS*. *SIZE* is the logarithm of market capitalization. *TOBINQ* is Tobin's Q. *ROA* is return on assets. *INT_CAP* is the ratio of capitalized intangible capital to assets, following (Peters and Taylor, 2017). *INTEGRITY*, *TEAMWORK*, *INNOVATION*, *RESPECT*, and *QUALITY* are measures of corporate culture from (Li et al., 2020), constructed using earnings calls transcripts. *BW* is an indicator for firms included in the Best Workplaces list compiled by Great Places to Work in America, obtained from Boustanifar and Kang (2022). All continuous explanatory variables are standardized to have mean zero and unit standard deviation. All regressions include a constant term, which is omitted for brevity. Standard errors, two-way clustered by industry and year, are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)
VARIABLES		PS		
SIZE	0.01	-0.01	-0.01	0.00
	(0.01)	(0.02)	(0.02)	(0.01)
TOBINQ	0.03**	0.03***	0.03**	0.02**
	(0.01)	(0.01)	(0.01)	(0.01)
ROA	-0.03	-0.05**	-0.04*	-0.04**
	(0.02)	(0.02)	(0.02)	(0.02)
INT_CAP	0.04*	0.02	0.03	0.02
	(0.02)	(0.02)	(0.02)	(0.02)
INTEGRITY		-0.03*	-0.03**	-0.03**
		(0.01)	(0.01)	(0.01)
TEAMWORK		0.05***	0.04***	0.03**
		(0.01)	(0.01)	(0.01)
INNOVATION		0.08***	0.07***	0.04***
		(0.01)	(0.01)	(0.01)
RESPECT		-0.00	-0.01	-0.01
		(0.01)	(0.01)	(0.01)
QUALITY		-0.02	-0.03	-0.04*
		(0.02)	(0.02)	(0.02)
BW		0.19***	0.22***	0.16***
		(0.05)	(0.04)	(0.03)
Observations	12,157	10,467	10,467	10,467
R-squared	0.01	0.03	0.04	0.05
Year FE	No	No	Yes	Yes
Industry FE	No	No	No	Yes

Table 3: Correlation between psychological safety and Glassdoor ratings

This table reports the correlations between various Glassdoor ratings and our measure of psychological safety, *PS*. *RATING* is the overall firm's rating in Glassdoor. *CULTURE* is the rating for culture and values. *WORKLIFE* is the rating for work-life balance. *COMPENSATION* is the rating for compensation and benefits. *CAREER* is the rating of career opportunities. *DIVERSITY* is the rating for diversity and inclusion, which was introduced by Glassdoor in 2020. All variables are standardized to have mean zero and unit standard deviation. All regressions include year and industry fixed effects as well as a constant term, which is omitted for brevity. Standard errors, two-way clustered by industry and year, are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES				PS		
RATING	0.56***	0.47***	0.47***	0.47***	0.47***	0.50**
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
CULTURE		0.14***	0.14***	0.23***	0.22***	0.28*
		(0.02)	(0.03)	(0.04)	(0.05)	(0.04)
WORKLIFE			0.00	0.06**	0.06*	0.11
			(0.02)	(0.03)	(0.03)	(0.05)
COMPENSATION				-0.18***	-0.19***	-0.20**
				(0.02)	(0.02)	(0.01)
CAREER					0.02	-0.09
					(0.02)	(0.06)
DIVERSITY						0.01
						(0.01)
Observations	14,353	13,504	13,504	13,504	13,504	3,377
R-squared	0.33	0.35	0.35	0.37	0.37	0.46

Table 4: Psychological safety and innovation

This table reports the relation between psychological safety (*PS*) and future patents. The dependent variable is the one-year-ahead logarithm of the number of granted patents. *SIZE* is the logarithm of market capitalization. *MB* is the logarithm of market-to-book ratio. *R&D* is the R&D expenditures scaled by market capitalization. *RATING* is the firm's overall employee rating from Glassdoor. *INTEGRITY*, *TEAMWORK*, *INNOVATION*, *RESPECT*, and *QUALITY* are measures of corporate culture from (Li et al., 2020). *BW* is an indicator for firms included in the Best Workplaces list compiled by Great Places to Work in America, obtained from Boustanifar and Kang (2022). *CULTURE* is the rating of culture and values from Glassdoor. All continuous right-hand-side variables are winsorized at the 1% and 99% levels and standardized. The constant term is omitted for brevity. Standard errors are two-way clustered by industry and year and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)
VARIABLES	f1.PATENTS			
PS	0.04***	0.04***	0.06***	0.04**
	(0.01)	(0.01)	(0.02)	(0.02)
SIZE	0.87***	1.14***	1.15***	1.14***
	(0.08)	(0.08)	(0.09)	(0.09)
MB	-0.04*	-0.05	-0.06	-0.05
	(0.02)	(0.03)	(0.04)	(0.04)
RD		0.42***	0.39***	0.40***
		(0.04)	(0.05)	(0.05)
RATING			-0.02	-0.03
			(0.03)	(0.06)
INTEGRITY			-0.02	-0.03
			(0.04)	(0.04)
TEAMWORK			0.11	0.11
			(0.07)	(0.07)
INNOVATION			0.08	0.09
			(0.05)	(0.06)
QUALITY			0.05	0.06
			(0.06)	(0.06)
RESPECT			-0.11**	-0.10**
			(0.05)	(0.05)
BW			0.55***	0.51***
			(0.15)	(0.16)
CULTURE				0.01
				(0.04)
Observations	5,065	3,766	3,509	3,225
R-squared	0.52	0.60	0.62	0.62
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes

Table 5: Psychological safety and current market valuation

This table reports the relationship between psychological safety (*PS*)and current market valuation, measured by the logarithm of market-to-book ratio. *PSYCH SAFETY* is psychological safety, constructed from employees' comments on Glassdoor. *RATING* is the overall employee rating/satisfaction of the firm on Glassdoor. *SIZE* is the logarithm of market capitalization. *LEVERAGE* is total debt to assets. *INT_CAP* is the ratio of capitalized intangible capital to assets, following (Peters and Taylor, 2017). All regressions include a constant term, which is omitted for brevity. Standard errors, two-way clustered by industry and year, are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)
VARIABLES		MB		
PS	-0.02	0.01	0.02	0.01
	(0.02)	(0.01)	(0.01)	(0.01)
RATING	0.14***	0.07**	0.05*	0.05**
	(0.04)	(0.03)	(0.02)	(0.02)
SIZE		0.38***	0.38***	0.36***
		(0.04)	(0.04)	(0.04)
LEVERAGE		0.07*	0.06	0.10***
		(0.04)	(0.04)	(0.03)
INT_CAP		0.42***	0.42***	0.24***
		(0.04)	(0.05)	(0.03)
Observations	13,498	13,340	13,340	13,340
R-squared	0.02	0.26	0.27	0.38
Year FE	No	No	Yes	Yes
Industry FE	No	No	No	Yes

Table 6: Risk-adjusted returns to psychological safety portfolios

This table reports the results of time series factor regressions for equal-weighted long–short portfolios formed based on psychological safety. Three different factor models are used: the five-factor model of Fama and French (2015) augmented with momentum (FF6); the q-factor model of Hou et al. (2021) (Q5); and the mispricing factor model of Stambaugh and Yuan (2017) (SY4). Newey—West standard errors are calculated and the associated t-statistics are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)
VARIABLES	FF6	Q5	SY4
α	5.18***	4.57***	5.99***
	(3.12)	(2.90)	(3.34)
MKT	-0.08**	-0.10**	-0.09**
	(-2.33)	(-2.57)	(-2.39)
SMB	0.01	0.10	-0.02
	(0.20)	(1.43)	(-0.29)
HML	0.01		
	(0.19)		
MOM	0.09**		
	(2.44)		
CMA	-0.34***		
	(-3.78)		
RMW	-0.04		
	(-0.48)		
qIA		-0.32***	
		(-4.07)	
qROE		0.02	
		(0.46)	
qEG		0.13	
		(1.49)	
MGMT			-0.38***
			(-5.55)
PERF			0.06
			(0.97)
Observations	114	114	114

Table 7: Risk-adjusted returns to psychological safety portfolios: Size-neutral portfolios

This table reports time-series factor-regression alphas for size-neutral equal- and value-weighted long–short portfolios formed on psychological safety. Size-neutral portfolios are constructed using independent annual double sorts into psychological-safety quintiles and size terciles using beginning-of-period June market equity, with portfolio returns averaged equally across size terciles. Panel A reports equal-weighted returns and Panel B value-weighted returns. Alphas are estimated using three factor models: the five-factor model of Fama and French (2015) augmented with momentum (FF6); the q-factor model of Hou et al. (2021) (Q5); and the mispricing factor model of Stambaugh and Yuan (2017) (SY4).Newey—West standard errors are used, and the corresponding t-statistics are reported in parentheses. The sample period spans July 2012 to December 2021. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	(1) FF6	(2) Q5	(3) SY4
Panel A: Equal-weighted			
α	3.71**	3.03**	4.53***
	(2.48)	(2.10)	(2.87)
Observations	114	114	114
Panel B: Value-weighted			
α	4.26***	3.85**	5.19***
	(2.73)	(2.60)	(3.14)
Observations	114	114	114

Table 8: Risk-adjusted returns to the short and long legs of psychological safety portfolios

This table reports the results of time-series factor regressions for the short (Low) and long (High) legs of portfolios formed on psychological safety. Three different factor models are used: the five-factor model of Fama and French (2015) augmented with momentum (FF6); the q-factor model of Hou et al. (2021) (Q5); and the mispricing factor model of Stambaugh and Yuan (2017) (SY4). Newey—West standard errors are used, and the corresponding t-statistics are reported in parentheses. The sample is from July 2012 to December 2021. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively..

	(1)	(2)	(3)	(4)	(5)	(6)
	FF6		Q5		SY4	
	Low	High	Low	High	Low	High
α	-0.38	4.80***	0.61	5.18***	-1.30	4.69***
	(-0.32)	(3.12)	(0.50)	(3.55)	(-1.01)	(2.84)
Observations	114	114	114	114	114	114

Table 9: Glassdoor ratings and abnormal returns

This table reports time-series factor regressions for long–short portfolios formed on firms' Glassdoor ratings and on residualized psychological safety. Panel A reports results for portfolios sorted on Glassdoor ratings. Panel B reports results for portfolios sorted on residualized psychological safety, defined as the residuals from a regression of psychological safety on Glassdoor ratings. Three factor models are used: the five-factor model of Fama and French (2015) augmented with momentum (FF6); the q-factor model of Hou et al. (2021) (Q5); and the mispricing factor model of Stambaugh and Yuan (2017) (SY4). Newey—West standard errors are calculated and the associated t-statistics are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

	(1) FF6	(2) Q5	(3) SY4
Panel A. Glassdoor rating-sorted portfolio			
α	0.58	-0.16	0.64
	(0.34)	(-0.09)	(0.38)
Observations	114	114	114
Panel B. Residualized psychological safety portfolio			
α	5.02**	5.08**	5.82***
	(2.56)	(2.42)	(2.75)
Observations	114	114	114

Table 10: Risk-adjusted returns to psychological safety portfolios: Controlling for intangibles-related factors

This table reports the results of time-series factor regressions for long-–short portfolios formed based on psychological safety. Column (1) uses the five-factor model of Fama and French (2015) augmented with momentum (FF6). Column (2) adds *ORGCAP*, an intangible (organizational capital) factor from Eisfeldt and Papanikolaou (2013). Column (3) includes an R&D factor constructed as R&D expenditures scaled by market capitalization and obtained from the Global Factor Data website. Column (4) includes both *ORGCAP* and the R&D factor. The sample period spans July 2012 to December 2021. Newey—West standard errors are used, and the corresponding t-statistics are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

VARIABLES	(1)	(2)	(3)	(4)
α	5.18***	5.18***	5.90***	5.50***
	(3.12)	(3.12)	(3.22)	(3.11)
MKT	-0.08**	-0.09**	-0.08**	-0.09**
	(-2.33)	(-2.49)	(-2.29)	(-2.45)
SMB	0.01	-0.06	0.01	-0.05
	(0.20)	(-1.00)	(0.10)	(-0.94)
HML	0.01	0.03	0.00	0.02
	(0.19)	(0.44)	(0.02)	(0.33)
MOM	0.09**	0.09**	0.07*	0.08**
	(2.44)	(2.53)	(1.87)	(2.25)
CMA	-0.34***	-0.36***	-0.36***	-0.37***
	(-3.78)	(-4.51)	(-4.25)	(-4.66)
RMW	-0.04	-0.07	-0.08	-0.09
	(-0.48)	(-0.94)	(-1.12)	(-1.14)
ORGCAP		0.21**		0.20*
		(2.02)		(1.75)
R&D			-0.09*	-0.04
			(-1.85)	(-0.77)
Observations	114	114	114	114

Table 11: Quarterly earnings surprise and market reaction

This table reports the relation between psychological safety (PS) and standardized unexpected quarterly earnings (SUE), as well as the corresponding market reaction to earnings announcements. The dependent variable in columns 1–4 is standardized unexpected quarterly earnings (SUE), while the dependent variable in columns 5–8 is cumulative abnormal returns (CAR) following the earnings call. $SIZE$ is the lagged logarithm of market capitalization, and BM is the lagged logarithm of the market-to-book ratio. All regressions include a constant term, which is not reported in the output. Standard errors are two-way clustered by industry and year, and the corresponding t-statistics are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	SUE				CAR			
PS	0.06***	0.06**	0.06**	0.06*	0.21**	0.16*	0.16**	0.20**
	(3.43)	(2.30)	(2.37)	(2.15)	(2.43)	(2.02)	(2.45)	(3.12)
SIZE		0.02**	0.02**	0.02**		-0.07	-0.06	-0.06
		(2.30)	(2.49)	(2.41)		(-0.75)	(-0.66)	(-0.68)
BM		0.03*	0.03	0.03		0.04	0.04	0.03
		(2.13)	(1.62)	(1.62)		(0.48)	(0.55)	(0.37)
Observations	43,313	41,134	41,134	41,134	43,221	41,037	41,037	41,037
R-squared	0.00	0.01	0.01	0.03	0.00	0.00	0.01	0.02
Year FE	No	Yes	Yes	No	No	Yes	Yes	No
Industry FE	No	No	Yes	No	No	No	Yes	No
Industry×Year FE	No	No	No	Yes	No	No	No	Yes

Table 12: Psychological safety and cross-sectional stock returns

This table reports the results of monthly cross-sectional Fama–MacBeth regressions. The dependent variable is the one-month-ahead stock return. PS is the aggregate measure of psychological safety constructed from all employee reviews. PS_BC and PS_WC denote measures of psychological safety constructed from blue-collar and white-collar employees, respectively. $RATING$ is the firm's Glassdoor rating, representing overall employee satisfaction. BW is an indicator for firms included in the Best Workplaces list compiled by Great Places to Work in America, obtained from Boustanifar and Kang (2022). $R\&D$ is research and development expenditures scaled by market capitalization. Firm characteristics included are beta, logarithm of market capitalization, logarithm of the book-to-market ratio, 12-month momentum, profitability, asset growth, trading volume, return skewness, the logarithm of stock price, leverage, share repurchases, illiquidity, and intangible capital from Peters and Taylor (2017) over assets. These firm characteristics are omitted from the reported output for brevity. All individual right-hand-side variables are winsorized at the 1% and 99% levels and standardized. Newey—West standard errors are used, and the corresponding t-statistics are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)	(5)
VARIABLES	f1.excess return				
PS	0.04	0.05	0.02		
	(1.25)	(0.97)	(0.52)		
PS × R&D			0.14**		
			(2.11)		
PS_WC				0.07*	0.06
				(1.83)	(1.41)
PS_WC × R&D					0.14**
					(2.13)
PS_BC				0.03	0.01
				(0.53)	(0.18)
RATING		0.00	0.01	-0.00	-0.00
		(0.02)	(0.09)	(-0.05)	(-0.02)
BW		0.23	0.19	0.21	0.18
		(1.14)	(0.97)	(1.07)	(0.92)
R&D			0.19		0.19
			(1.65)		(1.63)
Characteristics	Yes	Yes	Yes	Yes	Yes
Observations	113,397	113,397	113,397	113,397	113,397
R-squared	0.09	0.09	0.10	0.09	0.10

Appendix A Details on Measuring Psychological Safety

This appendix documents the procedure used to label Glassdoor employee reviews for psychological safety using a large language model and to extend these labels to the full review corpus using supervised learning.

A.1 Conceptual Motivation and Prompt Design

Psychological safety is a latent organizational attribute that is rarely stated explicitly in employee narratives. Instead, it is often conveyed indirectly through descriptions of leadership behavior, communication norms, feedback practices, and interpersonal dynamics. To capture this construct systematically, we design a structured prompt grounded in Edmondson's definition of psychological safety as employees' shared belief that they can engage in interpersonal risk-taking without fear of punishment, embarrassment, or negative consequences (Edmondson, 1999). The prompt instructs the model to assess whether the workplace environment described in a review supports behaviors such as speaking up with ideas or concerns, offering candid feedback to managers, admitting mistakes, asking questions, or expressing disagreement. Because psychological safety may be implied rather than directly stated, and because employees rarely use the terminology of organizational studies, the prompt emphasizes tone, context, and implication rather than relying solely on explicit language.

To operationalize the construct, the prompt defines a five-point ordinal scale ranging from clearly positive to clearly negative psychological safety. Reviews that provide explicit evidence of encouragement, open dialogue, welcomed dissent, or support after mistakes are coded as strongly positive. Reviews that imply a safe environment for speaking up and taking interpersonal risks are coded as weakly positive. Reviews that do not meaningfully address interpersonal dynamics, or that focus exclusively on compensation, hours, benefits, or logistical aspects of the job, are coded as neutral. Reviews that suggest fear of speaking up, criticizing decisions, or admitting mistakes are coded as weakly negative, while reviews that explicitly describe retaliation, punishment, public shaming, or other negative consequences for speaking up are coded as strongly negative.

The full prompt used in the labeling procedure is as follows:

> *You are analyzing Glassdoor employee reviews to assess psychological safety in the workplace. Psychological safety is the belief that individuals can speak up with ideas, concerns, questions, or mistakes without fear of punishment, embarrassment, or negative consequences. It reflects whether the work environment supports interpersonal risk-taking—such as raising concerns, offering candid feedback to*

managers, admitting errors, asking questions, or expressing disagreement—without fear of negative interpersonal consequences.

Scoring Scale:

+2 — Clearly positive: Explicit evidence of psychological safety.

+1 — Implicitly positive: Suggests psychological safety without saying it explicitly.

0 — Neutral / Not relevant: No clear information about psychological safety.

-1 — Implicitly negative: Suggests lack of psychological safety without saying it directly.

-2 — Clearly negative: Explicit absence of psychological safety.

Instructions:

Use the content of the review to assess interpersonal risk-taking and the social-emotional environment relevant for psychological safety.

Consider tone, implication, and context, not only explicit statements.

If the review is off-topic or only discusses pay, hours, perks, benefits, or logistics, assign a 0.

Return only the appropriate label: -2, -1, 0, +1, or +2—nothing else.

Now label the following review using only one number: -2, -1, 0, +1, or +2.

A.2 API-Based Implementation

We use GPT-4o to generate initial psychological safety labels for a sample of Glassdoor reviews. GPT-4o is well suited to this task because the construct often depends on contextual nuance, implication, and semantic relationships that may not be captured by keyword-based methods. We implement the prompt through the GPT-4o application programming interface.

Each Glassdoor review is processed independently. The full text of the review is submitted together with the prompt shown above, and the model returns a single numeric label corresponding to the five-point psychological safety scale. The instruction to return only a numerical value ensures standardized, machine-readable output and avoids ambiguity in downstream processing. Using this procedure, we label 50,000 reviews. In this setting, the language model serves as a theory-guided text coder that maps unstructured employee narratives into a standardized measure of psychological safety.

A.3 Integration with Supervised Learning

After generating labels with GPT-4o, we use these labels as training targets for a supervised machine learning model. The goal is to extend the labeling procedure beyond the initially labeled reviews and predict psychological safety for the remaining Glassdoor reviews. To do so, we fine-tune DeBERTa-v3-base, a pretrained transformer-based language model that can capture contextual and semantic relationships in text.

DeBERTa-v3-base is a transformer encoder model in the same broad class as BERT, but incorporates subsequent improvements in attention and pretraining. In particular, DeBERTa separates content and position information in the attention mechanism, and DeBERTa-v3 uses an ELECTRA-style replaced-token-detection objective, which improves pretraining efficiency and downstream natural-language-understanding performance. We use DeBERTa-v3-base rather than FinBERT because our input text consists of employee reviews rather than financial news or disclosures. FinBERT is designed for financial-domain sentiment classification, whereas our task requires learning a theory-guided organizational construct from workplace narratives. Fine-tuning DeBERTa-v3-base on GPT-4o-generated labels allows the classifier to adapt to the language of employee reviews and identify textual patterns associated with psychological safety.

The fine-tuning procedure proceeds as follows. First, each review is tokenized, converting the raw text into numerical input representations that can be processed by the model. Tokenization allows the model to handle both words and subword units, and reviews are padded or truncated to a fixed maximum sequence length to ensure consistent input dimensions. The tokenized review is then passed through the transformer model, which produces contextual embeddings that represent the meaning of words and phrases in relation to the surrounding text.

To generate predictions, we attach a classification head to the transformer output. The classification head maps the review representation into probabilities over the five psychological safety classes, corresponding to labels from -2 to $+2$. These probabilities are normalized using a softmax function, and the predicted class is the label with the highest probability. During training, model parameters are updated to minimize cross-entropy loss between the predicted probabilities and the GPT-4o-generated labels. We use the AdamW optimizer and train the model over multiple epochs, allowing it to learn textual patterns associated with different levels of psychological safety and apply them to reviews outside the initial labeled sample.

A.4 Model Application

After fine-tuning, we apply the DeBERTa-v3-base model to the remaining Glassdoor reviews to generate review-level psychological safety scores. Each review is tokenized and processed through the model, which outputs a predicted label on the five-point scale (-2, -1, 0, $+1$, $+2$). This step extends the GPT-4o-based labeling procedure to the full review corpus, allowing the model to identify both explicit and implicit cues of psychological safety in unstructured employee text.

Appendix Tables

Table A1: Psychological safety and innovation with industry×year fixed effects

This table reports the relation between psychological safety (*PS*) and future patents. The dependent variable is the one-year-ahead logarithm of the number of granted patents. *SIZE* is the logarithm of market capitalization. *MB* is the logarithm of market-to-book ratio. *R&D* is the R&D expenditures scaled by market capitalization. *RATING* is the firm's overall employee rating from Glassdoor. *INTEGRITY*, *TEAMWORK*, *INNOVATION*, *RESPECT*, and *QUALITY* are measures of corporate culture from (Li et al., 2020). *BW* is an indicator for firms included in the Best Workplaces list compiled by Great Places to Work in America, obtained from Boustanifar and Kang (2022). *CULTURE* is the rating of culture and values from Glassdoor. All right-hand-side variables are winsorized at the 1% and 99% levels and standardized. The constant term is omitted for brevity. Standard errors are two-way clustered at the industry and year level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)
VARIABLES		f1.PATENTS		
PS	0.05***	0.04***	0.05**	0.04**
	(0.01)	(0.01)	(0.02)	(0.02)
SIZE	0.89***	1.15***	1.16***	1.15***
	(0.08)	(0.08)	(0.09)	(0.09)
MB	-0.03	-0.05	-0.05	-0.05
	(0.02)	(0.03)	(0.04)	(0.04)
RD		0.43***	0.40***	0.41***
		(0.05)	(0.05)	(0.05)
RATING			-0.02	-0.03
			(0.04)	(0.06)
INTEGRITY			-0.01	-0.02
			(0.04)	(0.05)
TEAMWORK			0.09	0.09
			(0.07)	(0.07)
INNOVATION			0.09	0.09
			(0.06)	(0.06)
QUALITY			0.07	0.08
			(0.06)	(0.06)
RESPECT			-0.10*	-0.10*
			(0.05)	(0.05)
BW			0.49***	0.47***
			(0.15)	(0.16)
CULTURE				-0.01
				(0.04)
Observations	5,019	3,717	3,459	3,178
R-squared	0.56	0.63	0.64	0.64
Industry×Year FE	Yes	Yes	Yes	Yes

Table A2: Psychological safety and innovation: Poisson regressions

This table reports the relation between psychological safety (*PS*) and future patents using Poisson regressions. The dependent variable is the one-year-ahead number of granted patents. *SIZE* is the logarithm of market capitalization. *MB* is the logarithm of market-to-book ratio. *R&D* is the R&D expenditures scaled by market capitalization. *RATING* is the firm's overall employee rating from Glassdoor. *INTEGRITY*, *TEAMWORK*, *INNOVATION*, *RESPECT*, and *QUALITY* are measures of corporate culture from (Li et al., 2020). *BW* is an indicator for firms included in the Best Workplaces list compiled by Great Places to Work in America, obtained from Boustanifar and Kang (2022). *CULTURE* is the rating of culture and values from Glassdoor. All continuous right-hand-side variables are winsorized at the 1% and 99% levels and standardized. The constant term is omitted for brevity. Standard errors are two-way clustered at the industry and year level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)
VARIABLES	f1.PATENTS			
PS	0.34***	0.33**	0.36**	0.35**
	(0.13)	(0.14)	(0.14)	(0.15)
SIZE	2.33***	2.37***	2.38***	2.40***
	(0.23)	(0.24)	(0.24)	(0.24)
MB	0.02	0.02	0.01	0.02
	(0.05)	(0.05)	(0.05)	(0.06)
RD	0.71***	0.71***	0.73***	0.72***
	(0.11)	(0.12)	(0.12)	(0.13)
RATING	-0.53**	-0.53*	-0.64**	-0.61**
	(0.27)	(0.29)	(0.30)	(0.30)
INTEGRITY	-0.06	-0.10	-0.07	-0.11
	(0.08)	(0.09)	(0.09)	(0.10)
TEAMWORK	0.01	0.04	0.02	0.05
	(0.11)	(0.12)	(0.12)	(0.13)
INNOVATION	-0.01	-0.02	-0.01	-0.02
	(0.11)	(0.12)	(0.12)	(0.13)
QUALITY	0.01	-0.01	0.00	-0.01
	(0.04)	(0.04)	(0.04)	(0.04)
RESPECT	-0.67***	-0.69***	-0.66***	-0.68***
	(0.19)	(0.19)	(0.19)	(0.19)
BW	-0.24**	-0.32***	-0.21**	-0.30***
	(0.10)	(0.11)	(0.10)	(0.11)
CULTURE			0.06	0.04
			(0.04)	(0.06)
Observations	4,199	4,139	3,811	3,755
Year FE	Yes	No	Yes	No
Industry FE	Yes	No	Yes	No
Industry×Year FE	No	Yes	No	Yes

Table A3: Psychological safety and patent citations

This table reports the relation between psychological safety (*PS*) and citations to patents using Poisson regressions. The dependent variable is the one-year-ahead number of citations. *SIZE* is the logarithm of market capitalization. *MB* is the logarithm of market-to-book ratio. *R&D* is the R&D expenditures scaled by market capitalization. *RATING* is the firm's overall employee rating from Glassdoor. *INTEGRITY*, *TEAMWORK*, *INNOVATION*, *RESPECT*, and *QUALITY* are measures of corporate culture from (Li et al., 2020). *BW* is an indicator for firms included in the Best Workplaces list compiled by Great Places to Work in America, obtained from Boustanifar and Kang (2022). *CULTURE* is the rating of culture and values from Glassdoor. All continuous right-hand-side variables are winsorized at the 1% and 99% levels and standardized. The constant term is omitted for brevity. Standard errors are two-way clustered at the industry and year level and reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)
VARIABLES	f1.CITATIONS			
PS	-0.16	-0.19	-0.18	-0.22
	(0.15)	(0.15)	(0.16)	(0.16)
SIZE	2.21***	2.21***	2.21***	2.21***
	(0.18)	(0.18)	(0.18)	(0.18)
MB	-0.00	-0.00	-0.01	-0.01
	(0.05)	(0.06)	(0.05)	(0.07)
RD	0.66***	0.67***	0.67***	0.68***
	(0.14)	(0.15)	(0.14)	(0.15)
RATING	0.07	0.13	0.02	0.05
	(0.10)	(0.11)	(0.08)	(0.10)
INTEGRITY	-0.10	-0.10	-0.09	-0.09
	(0.08)	(0.09)	(0.08)	(0.09)
TEAMWORK	0.09	0.10	0.10	0.11
	(0.12)	(0.13)	(0.12)	(0.13)
INNOVATION	-0.04	-0.02	-0.04	-0.03
	(0.14)	(0.14)	(0.14)	(0.14)
QUALITY	0.14*	0.15**	0.14**	0.16**
	(0.07)	(0.07)	(0.07)	(0.07)
RESPECT	-0.39	-0.42*	-0.39	-0.42*
	(0.24)	(0.23)	(0.24)	(0.23)
BW	-0.34*	-0.46**	-0.34	-0.48**
	(0.18)	(0.22)	(0.23)	(0.23)
CULTURE			0.08	0.11
			(0.10)	(0.09)
Observations	4,191	4,114	3,804	3,747
Year FE	Yes	No	Yes	No
Industry FE	Yes	No	Yes	No
Industry×Year FE	No	Yes	No	Yes

Table A4: Psychological safety and current market valuation measured by price-to-earnings

This table reports the relationship between psychological safety (*PS*) and current market valuation, measured by the logarithm of price-to-earnings ratio. *PSYCH SAFETY* is psychological safety, constructed from employees' comments on Glassdoor. *RATING* is the overall employee rating/satisfaction of the firm on Glassdoor. *SIZE* is the logarithm of market capitalization. *LEVERAGE* is total debt to assets. *INT_CAP* is the ratio of capitalized intangible capital to assets, following (Peters and Taylor, 2017). All regressions include a constant term, which is omitted for brevity. Standard errors, two-way clustered by industry and year, are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

	(1)	(2)	(3)	(4)
VARIABLES		PE		
PS	0.01	0.01	0.01	0.00
	(0.02)	(0.02)	(0.02)	(0.01)
RATING	0.05	0.03	0.01	-0.01
	(0.03)	(0.03)	(0.02)	(0.02)
SIZE		0.11***	0.11***	0.09**
		(0.03)	(0.03)	(0.03)
LEVERAGE		-0.09*	-0.10**	-0.08**
		(0.04)	(0.04)	(0.03)
INT_CAP		0.18**	0.17**	0.10
		(0.06)	(0.06)	(0.06)
Observations	8,357	8,319	8,319	8,318
R-squared	0.00	0.04	0.06	0.13
Year FE	No	No	Yes	Yes
Industry FE	No	No	No	Yes